🎶 Hit Song Predictor: Amapiano & Afrobeats 🎧

This project explores whether a song's audio features and metadata can predict its hit potential. We focus on Amapiano and Afrobeats, genres currently dominating African and global music scenes.

In [2]:
```python
from IPython.display import display, HTML
display(HTML("<p><em> Press play to start the mix if your browser blocked autoplay.</em></p>"))

#Playlist intro
display(HTML("""
<h3>🎧 Now Playing: Amapiano & Afrobeats Mix</h3>
<p><strong>Tracklist:</strong><br>
1. Jealousy – Khalil Harrison (Amapiano)<br>
2. Shake Ah – Tyla (Amapiano)<br>
3. Woman – Rema (Afrobeats)<br>
4. Zenzele – Uncle Waffles (Amapiano)<br>
5. No Competition – Davido ft. Asake (Afrobeats)<br>
6. Bunda – Musa Keys (Amapiano)<br>
7. Laho – Shalipopi (Afrobeats)<br>
8. My Darling – Chella (Afrobeats)
</p>
"""))

#Audio player (autoplay with user controls)
display(HTML("""
<audio autoplay controls>
  <source src="Afrobeats and Amapiano mix.mp3" type="audio/mpeg">
  Your browser does not support the audio element.
</audio>
"""))
```

Press play to start the mix if your browser blocked autoplay.

🎧 Now Playing: Amapiano & Afrobeats Mix

Tracklist:

1. Jealousy – Khalil Harrison (Amapiano)
2. Shake Ah – Tyla (Amapiano)
3. Woman – Rema (Afrobeats)
4. Zenzele – Uncle Waffles (Amapiano)
5. No Competition – Davido ft. Asake (Afrobeats)
6. Bunda – Musa Keys (Amapiano)
7. Laho – Shalipopi (Afrobeats)
8. My Darling – Chella (Afrobeats)

0:00 / 10:24

Purpose of Project

Applying machine learning to music has proven insightful: computers can detect qualities of songs that resonate with audiences. Academic resemarch has shown that hit songs often share certain characteristics. For example, a study analyzing 30 years of music found “successful songs are happier, brighter, more party-like, more danceable and less sad than most songs” (Kaplan, 2018).In summary, building a hit song prediction model is important and appealing because it combines scientific rigor with cultural relevance. It helps company stakeholders make smarter decisions.

1. Import Libraries and Set Up

In [6]:
```
#Import necessary libraries for Spotify API access, data handling, and file management
import spotipy
from spotipy.oauth2 import SpotifyOAuth
import pandas as pd
import time
import os
```

```
#Create a directory called "data" to save all downloaded or processed datasets
#This keeps the project organized and avoids cluttering the main directory
os.makedirs("data", exist_ok=True)
```

2. Spotify API Authentication

In [8]:
```
#Authenticate with Spotify's API using OAuth 2.0 flow
#This gives access to user-specific data like saved tracks and private playlists
sp = spotipy.Spotify(auth_manager=SpotifyOAuth(
    client_id="053a18fb9a87491d8335598e56799c5e",
    client_secret= "",
    redirect_uri="http://127.0.0.1:8891/callback",
    scope="playlist-read-private playlist-read-collaborative"
))

#Test connection by pulling a few liked songs from Spotify account
#This confirms that access token works and successfully authenticatedresults = sp.current_user_saved_tracks(limit=5)
results = sp.current_user_saved_tracks(limit=5)
for item in results['items']:
    track = item['track']
    print(f"{track['name']} by {track['artists'][0]['name']}")
```

```
GIRLFRIEND by Tayc
For The Night by Tayo J
Petit génie by Jungeli
Focus On Me (All The Sexy Girls In The Club) by Darkoo
That Should Be Me by Justin Bieber
```

3. Extract Playlists Data from Spotify (looking for popularity score)

In [10]:
```
#Define a dictionary of confirmed working playlists across Afrobeats and Amapiano genres
#Each playlist maps to its unique Spotify ID for reliable API access
working_playlists = {
    'Best of Afrobeats 2025': '5FDBAbJobJWaKh1RDiqtyn',
    'TikTok Naija': '1H4Ws8FYRXbUCFgpYAZdK3',
    'Afrobeats Party': '1U8HSDxH8lXHQ38epJngtG',
    'Afrobeat 2025': '7IfWkPjxjtGpHKzvbZd8YV',
    'Weekly Top Grooves': '4z8jM6c6NLp0H0szju3flc',
    'African Heat': '3Uj9Y8xviyC4BvC9R49reX',
    'Best of Amapiano': '1JtkyBr4Is4ni1UQAa9AVg',
```

```
    'AmaPiano March-April 2025': '74QpABGrU1VAdBlnaJqATL',
    'Amapiano New Finds': '7LQ1WmcPCKJtXT5uQl3waU',
    'playlist_id' : '5sFUBxyx9qpGDFOCPBEd82'


}



#Authenticate with the Spotify API using secure OAuth credentials
#Scope is limited to playlist access for data collection purposes
sp = spotipy.Spotify(auth_manager=SpotifyOAuth(
    client_id="053a18fb9a87491d8335598e56799c5e",
    client_secret= "",
    redirect_uri="http://127.0.0.1:8891/callback",
    scope="playlist-read-private playlist-read-collaborative"
))

#Initialize container to hold track-level data collected from all playlists
track_data = []

#Loop through each playlist, fetch up to 100 tracks at a time, and append relevant metadata
for playlist_name, playlist_id in working_playlists.items():
    try:
        print(f"Fetching from: {playlist_name}")
        offset = 0
        while True:
            results = sp.playlist_items(playlist_id, limit=100, offset=offset)
            if not results['items']:
                break

            for item in results['items']:
                track = item['track']
                if track:  # Make sure it's not None
                    track_data.append({
                        'playlist': playlist_name,
                        'track_name': track['name'],
                        'artist': track['artists'][0]['name'],
                        'track_id': track['id'],
                        'popularity': track['popularity']
                    })
            offset += 100
            time.sleep(0.5)  #To respect rate limits
    except Exception as e:
```

```
        print(f"Failed to fetch from {playlist_name}: {e}")

#Convert collected data to a structured DataFrame and export to CSV for further analysis
df_tracks = pd.DataFrame(track_data)
df_tracks.to_csv("playlist_with_all_audio_features_complete (3).csv", index=False)
```

```
Fetching from: Best of Afrobeats 2025
Fetching from: TikTok Naija
Fetching from: Afrobeats Party
Fetching from: Afrobeat 2025
Fetching from: Weekly Top Grooves
Fetching from: African Heat
Fetching from: Best of Amapiano
Fetching from: AmaPiano March-April 2025
Fetching from: Amapiano New Finds
Fetching from: playlist_id
```

4. Collect Spotify Stream Data for Recent Trending Afrobeats and Amampiano Songs (Will Measure Popularity by Streams Per Day Later)

In [12]:

```
# Stream Data
#📊 Contains total stream counts, genre labels, and release dates for Afrobeats and Amapiano songs
#Data includes music from Nigeria (afrobeats top 30 current spotify songs) and South Africa (top 30 current spotify s

stream_data = {
    # --- AFROBEATS songs ---
    "my darling": {"streams": 4_924_774, "genre": "Afrobeats", "release_date": "2025-03-27"},
    "arike": {"streams": 22_434_165, "genre": "Afrobeats", "release_date": "2025-02-15"},
    "are you there?": {"streams": 31_946_737, "genre": "Afrobeats", "release_date": "2024-08-16"},
    "awolowo": {"streams": 48_842_828, "genre": "Afrobeats", "release_date": "2024-08-14"},
    "beamer": {"streams": 11_580_503, "genre": "Afrobeats", "release_date": "2024-10-14"},
    "be there still": {"streams": 6_737_818, "genre": "Afrobeats", "release_date": "2025-02-14"},
    "chandelier": {"streams": 17_369_585, "genre": "Afrobeats", "release_date": "2024-11-21"},
    "na scra": {"streams": 7_248_693, "genre": "Afrobeats", "release_date": "2025-03-07"},
    "doha": {"streams": 27_380_251, "genre": "Afrobeats", "release_date": "2024-07-13"},
    "free of charge": {"streams": 1_915_678, "genre": "Afrobeats", "release_date": "2025-03-27"},
    "pity this boy (with victony)": {"streams": 21_224_424, "genre": "Afrobeats", "release_date": "2025-02-28"},
    "get better": {"streams": 4_857_075, "genre": "Afrobeats", "release_date": "2025-03-21"},
    "funds (feat. odumodublvck & chike)": {"streams": 42_455_477, "genre": "Afrobeats", "release_date": "2024-12-06"}
    "happy": {"streams": 8_112_951, "genre": "Afrobeats", "release_date": "2025-02-20"},
    "hey jago": {"streams": 2_993_456, "genre": "Afrobeats", "release_date": "2025-03-18"},
    "joy is coming": {"streams": 37_092_308, "genre": "Afrobeats", "release_date": "2024-12-18"},
```

```
    "JUJU (feat. Shallipopi)": {"streams": 41_465_826, "genre": "Afrobeats", "release_date": "2024-08-22"},
    "kese (dance)": {"streams": 45_694_575, "genre": "Afrobeats", "release_date": "2024-10-15"},
    "baby (is it a crime)": {"streams": 35_551_585, "genre": "Afrobeats", "release_date": "2025-02-07"},
    "shaolin": {"streams": 12_715_482, "genre": "Afrobeats", "release_date": "2025-02-10"},
    "management": {"streams": 6_204_585, "genre": "Afrobeats", "release_date": "2025-01-25"},
    "most wanted": {"streams": 1_837_793, "genre": "Afrobeats", "release_date": "2025-04-04"},
    "mario kart": {"streams": 8_687_810, "genre": "Afrobeats", "release_date": "2025-02-20"},
    "legolas": {"streams": 1_955_624, "genre": "Afrobeats", "release_date": "2025-03-31"},
    "trenches luv": {"streams": 4_312_356, "genre": "Afrobeats", "release_date": "2025-02-13"},
    "toy girl (with juno & valentino rose)": {"streams": 1_197_925, "genre": "Afrobeats", "release_date": "2025-03-31
    "venus": {"streams": 26_775_718, "genre": "Afrobeats", "release_date": "2025-02-21"},
    "why love": {"streams": 10_384_733, "genre": "Afrobeats", "release_date": "2025-02-12"},

    # --- AMAPIANO songs ---
    "Sdudla or Slender": {"streams": 2_894_712, "genre": "Amapiano", "release_date": "2025-03-20"},
    "Vuma Dlozi Lami (feat. Ancestral Rituals)": {"streams": 9_243_178, "genre": "Amapiano", "release_date": "2024-09
    "Ngisakuthanda": {"streams": 7_241_847, "genre": "Amapiano", "release_date": "2024-09-06"},
    "Ngibolekeni (feat. Seun1401, LeeMcKrazy, Blxckie, Pcee, Madumane & Kabelo Sings)": {"streams": 5_852_759, "genre
    "Vuka (feat. Thukuthela)": {"streams": 8_720_251, "genre": "Amapiano", "release_date": "2024-12-15"},
    "Uyaphapha Marn (feat. Scotts Maphuma & Kabelo Sings)": {"streams": 4_838_765, "genre": "Amapiano", "release_date
    "Wayengenalutho": {"streams": 4_610_909, "genre": "Amapiano", "release_date": "2025-02-21"},
    "Sohlala Sisonke": {"streams": 12_909_636, "genre": "Amapiano", "release_date": "2025-02-14"},
    "Bo Gogo (feat. Tracy & Thatohatsi)": {"streams": 6_078_812, "genre": "Amapiano", "release_date": "2025-01-31"},
    "HAUSAPIANO - Remix": {"streams": 21_129_492, "genre": "Amapiano", "release_date": "2024-10-31"},
    "Uvume Kanjani?": {"streams": 1_289_020, "genre": "Amapiano", "release_date": "2025-03-21"},
    "Biri Marung (feat. Sje Konka, Focalistic, DJ Maphorisa, Scotts Maphuma & CowBoii)": {"streams": 23_859_597, "gen
    "Romeo & Juliet": {"streams": 4_674_070, "genre": "Amapiano", "release_date": "2025-01-22"},
    "Shapa Bell": {"streams": 1_339_189, "genre": "Amapiano", "release_date": "2025-04-01"},
    "Abantwana Bakho (feat. Thatohatsi, Young Stunna & Nkosazana Daughter)": {"streams": 790_174, "genre": "Amapiano"
    "Malunde (feat. Springle)": {"streams": 864_570, "genre": "Amapiano", "release_date": "2024-12-13"},
    "Vulani (feat. Thatohatsi & Tracy)": {"streams": 3_295_409, "genre": "Amapiano", "release_date": "2024-12-09"},
    "Skuta Baba - Remix": {"streams": 7_294_010, "genre": "Amapiano", "release_date": "2024-12-06"},
    "Ungangilimazi (feat. Frank Mabeat)": {"streams": 4_937_736, "genre": "Amapiano", "release_date": "2024-09-20"},
    "All My Life": {"streams": 10_706_356, "genre": "Amapiano", "release_date": "2024-08-30"},
    "Awuhlabe Kabili": {"streams": 3_409_948, "genre": "Amapiano", "release_date": "2024-12-06"},
    "ZENZELE (feat. Royal MusiQ, Uncool MC, Xduppy, & CowBoii)": {"streams": 2_014_490, "genre": "Amapiano", "release_
    "Naledi": {"streams": 1_401_369, "genre": "Amapiano", "release_date": "2025-03-10"},
    "UYAH! (feat. 2wo Bunnies, Jay Music, & Imbongi Yosizi)": {"streams": 2_223_683, "genre": "Amapiano", "release_da
    "Ngiyakuthanda": {"streams": 3_686_853, "genre": "Amapiano", "release_date": "2025-02-02"},
    "Shayi'Moto (feat. Seemah & Yanda Woods)": {"streams": 9_941_374, "genre": "Amapiano", "release_date": "2024-11-0
    "Wishi Wishi (feat. Scotts Maphuma & Young Stunna)": {"streams":13_211_203, "genre": "Amapiano", "release_date":
    "Dear Ex Yami": {"streams": 5_231_036, "genre": "Amapiano", "release_date": "2024-09-22"},
    "Ama Gear": {"streams": 11_286_723, "genre": "Amapiano", "release_date": "2023-12-01"},
```

```
    "Kabza Chant (feat. Young Stunna, Nkosazana Daughter, Mthunzi, Nokwazi, Anzo, Mashudu, Murumba Pitch & Tman Xpres
    "Abo Nokthula (feat. The Exclusive SA, Scotts Maphuma, Kabelo Sings, Bontle Smith, 2woshort & Stompiiey)": {"stre

    # --- SONG CROSSING BOTH REGIONS (Afrobeats, charted in both NG & SA) ---
    "laho": {"streams": 24_981_448, "genre": "Afrobeats", "release_date": "2025-02-21"}  # Use the higher NG count
}
```

5. Extract Tiktok Virality (video uses) Data of Recent Afrobeats and Amapiano Music

In [14]:

```
#TikTok virality classification:
#TikTok virality is calculated as: TikTok uses / days since release, a song is considered a hit if its score is great
#This rule applies to both Afrobeats and Amapiano songs.

tiktok_trending = {
    # --- AFROBEATS ---
    "Laho": 1,
    "My Darling": 1,
    "Arike": 1,
    "Are you there?": 0,
    "Awolowo": 0,
    "Beamer": 0,
    "Be There Still": 0,
    "Chandelier": 0,
    "Na Scra": 1,
    "Doha": 0,
    "Free of Charge": 0,
    "PITY THIS BOY (with Victony)": 0,
    "Get Better": 0,
    "Funds (feat. ODUMODUBLVCK & Chike)": 1,
    "Happy": 1,
    "hey jago": 1,
    "Joy Is Coming": 1,
    "JUJU (feat. Shallipopi)": 1,
    "Kese (Dance)": 1,
    "Baby (Is it a Crime)": 1,
    "Shaolin": 1,
    "Management (with BIGKHALID)": 0,
    "Most Wanted": 0,
    "MARIO KART": 1,
    "LEGOLAS": 0,
    "Trenches Luv": 1,
```

```
    "TOY GIRL (with Juno & Valentino Rose)": 0,
    "Venus": 1,
    "WHY LOVE": 1,

    # --- AMAPIANO ---
    "Sdudla or Slender": 1,
    "Vuma Dlozi Lami (feat. Ancestral Rituals)": 0,
    "Ngisakuthanda": 0,
    "Ngibolekeni (feat. Seun1401, LeeMcKrazy, Blxckie, Pcee, Madumane & Kabelo Sings)": 1,
    "Vuka (feat. Thukuthela)": 1,
    "Uyaphapha Marn (feat. Scotts Maphuma & Kabelo Sings)": 1,
    "Wayengenalutho": 0,
    "Sohlala Sisonke": 1,
    "Bo Gogo (feat. Tracy & Thatohatsi)": 1,
    "HAUSAPIANO - Remix": 1,
    "Uvume Kanjani?": 1,
    "Biri Marung (feat. Sje Konka, Focalistic, DJ Maphorisa, Scotts Maphuma & CowBoii)": 1,
    "Romeo & Juliet": 1,
    "Shapa Bell": 0,
    "Abantwana Bakho (feat. Thatohatsi, Young Stunna & Nkosazana Daughter)": 0,
    "Malunde (feat. Springle)": 0,
    "Vulani (feat. Thatohatsi & Tracy)": 0,
    "Skuta Baba - Remix": 1,
    "Ungangilimazi (feat. Frank Mabeat)": 0,
    "All My Life": 0,
    "Awuhlabe Kabili": 0,
    "ZENZELE (feat. Royal MusiQ, Uncool MC, Xduppy, & CowBoii)": 0,
    "Naledi (w/ Naledi Aphiwe)": 0,
    "UYAH! (feat. 2wo Bunnies, Jay Music, & Imbongi Yosizi)": 0,
    "Ngiyakuthanda": 0,
    "Shayi'Moto (feat. Seemah & Yanda Woods)": 1,
    "Wishi Wishi (feat. Scotts Maphuma & Young Stunna)": 1,
    "Dear Ex Yami": 1,
    "Ama Gear": 0,
    "Kabza Chant (feat. Young Stunna, Nkosazana Daughter, Mthunzi, Nokwazi, Anzo, Mashudu, Murumba Pitch & Tman Xpres
    "Abo Nokthula (feat. The Exclusive SA, Scotts Maphuma, Kabelo Sings, Bontle Smith, 2woshort & Stompiiey)": 0
}
```

In [15]:
```
#Create DataFrame just with song names
df = pd.DataFrame({'song': list(tiktok_trending.keys())})
df['viral_on_tiktok'] = df['song'].map(tiktok_trending).fillna(0)
```

6. Observe Billboard Afrobeat Songs and Flag Recent Trending Songs

In [17]:
```
#Import fuzzy matching
from fuzzywuzzy import fuzz

#Billboard Africa Titles (as is)
billboard_africa_titles = [
    "Push 2 Start", "Water", "Move", "Baby (Is it a Crime)", "Shake It To The Max FLY", "Laho", "Get Better", "Piece 
    "Why Love", "Update", "Arike", "Be There Still", "Funds", "Joy Is Coming", "PITY THIS BOY with Victony", "SHAOLIN
    "Na Scra", "Bad For You", "Kese Dance", "Awake", "Mario Kart", "Trenches Luv", "Hey Jago", "Happy",
    "Good Vibes", "Bad Girl", "Who Does That", "Introduction", "Wetego", "Only Fans", "Taxi Driver", "New Taker",
    "Macho", "Apres Minuit", "Obimo", "iToro", "Panic", "Louder", "Bend", "JayJay", "Chandelier", "Beamer",
    "Going Intro", "Toma Toma", "Break Me Down", "A Million Blessings", "World Best", "lololufe"
]

import re

#Function to clean song titles by removing non-alphanumeric characters
def clean_title(title):
    return re.sub(r'[^a-zA-Z0-9]', '', title.lower().strip())

#Cleaned version of Billboard titles
billboard_africa_clean = [clean_title(title) for title in billboard_africa_titles]

#Clean your actual dataframe's song titles (adjust 'song' to your column name if different)
df['song_clean'] = df['song'].apply(clean_title)

from fuzzywuzzy import fuzz

#Function to fuzzily match (akin to a confidence score) song to the Billboard list
def is_billboard_hit(song, threshold=80):
    for bb_song in billboard_africa_clean:
        score = fuzz.token_sort_ratio(song, bb_song)
        if score >= threshold:
            return True
    return False

#Flag songs that appear in Billboard Africa chart
df['in_billboard_africa'] = df['song_clean'].apply(lambda x: 1 if is_billboard_hit(x) else 0)
```

In [18]:
```
#Assign 'Afrobeats' genre only for songs matched to Billboard
df.loc[df['in_billboard_africa'] == 1, 'genre'] = 'Afrobeats'


#Preview Billboard Africa chart hits with genre column
billboard_hits = df[df['in_billboard_africa'] == 1][['song', 'in_billboard_africa', 'genre']]
billboard_hits
```

Out[18]:

	song	in_billboard_africa	genre
0	Laho	1	Afrobeats
2	Arike	1	Afrobeats
5	Beamer	1	Afrobeats
6	Be There Still	1	Afrobeats
7	Chandelier	1	Afrobeats
8	Na Scra	1	Afrobeats
11	PITY THIS BOY (with Victony)	1	Afrobeats
12	Get Better	1	Afrobeats
14	Happy	1	Afrobeats
15	hey jago	1	Afrobeats
16	Joy Is Coming	1	Afrobeats
18	Kese (Dance)	1	Afrobeats
19	Baby (Is it a Crime)	1	Afrobeats
20	Shaolin	1	Afrobeats
23	MARIO KART	1	Afrobeats
25	Trenches Luv	1	Afrobeats
28	WHY LOVE	1	Afrobeats

7. Defining What Makes a 'hit' in Each Category

In [20]:
```
import pandas as pd
from rapidfuzz import fuzz
from IPython.display import display

#Load your playlist CSV
df_tracks = pd.read_csv("playlist_with_all_audio_features_complete (3).csv")

#Complete genre mapping
target_songs = {
    # --- AFROBEATS ---
    "My Darling": "Afrobeats",
    "Arike": "Afrobeats",
    "Are you there?": "Afrobeats",
    "Awolowo": "Afrobeats",
    "Beamer": "Afrobeats",
    "Be There Still": "Afrobeats",
    "Chandelier": "Afrobeats",
    "Na Scra": "Afrobeats",
    "Doha": "Afrobeats",
    "Free of Charge": "Afrobeats",
    "PITY THIS BOY (with Victony)": "Afrobeats",
    "Get Better": "Afrobeats",
    "Funds (feat. ODUMODUBLVCK & Chike)": "Afrobeats",
    "Happy": "Afrobeats",
    "hey jago": "Afrobeats",
    "Joy Is Coming": "Afrobeats",
    "JUJU (feat. Shallipopi)": "Afrobeats",
    "Kese (Dance)": "Afrobeats",
    "Baby (Is it a Crime)": "Afrobeats",
    "Shaolin": "Afrobeats",
    "Management": "Afrobeats",
    "Most Wanted": "Afrobeats",
    "MARIO KART": "Afrobeats",
    "LEGOLAS": "Afrobeats",
    "Trenches Luv": "Afrobeats",
    "TOY GIRL (with Juno, Valentino Rose)": "Afrobeats",
    "Venus": "Afrobeats",
    "WHY LOVE": "Afrobeats",
    "Laho": "Afrobeats",

    # --- AMAPIANO ---
    "Sdudla or Slender": "Amapiano",
    "Vuma Dlozi Lami (feat. Ancestral Rituals)": "Amapiano",
```

```
    "Ngisakuthanda": "Amapiano",
    "Ngibolekeni (feat. Seun1401, LeeMcKrazy, Blxckie, Pcee, Madumane & Kabelo Sings)": "Amapiano",
    "Vuka (feat. Thukuthela)": "Amapiano",
    "Uyaphapha Marn (feat. Scotts Maphuma...)": "Amapiano",
    "Wayengenalutho": "Amapiano",
    "Sohlala Sisonke": "Amapiano",
    "Bo Gogo (feat. Tracy & Thathohatsi)": "Amapiano",
    "HAUSAPIANO - Remix": "Amapiano",
    "Uvume Kanjani?": "Amapiano",
    "Biri Marung (feat. Sje Konka, Focalistic, DJ Maphorisa, Scotts Maphuma & CowBoii)": "Amapiano",
    "Romeo & Juliet": "Amapiano",
    "Shapa Bell": "Amapiano",
    "Abantwana Bakho (feat. Thatohatsi, Young Stunna & Nkosazana Daughter)": "Amapiano",
    "Malunde (feat. Springle)": "Amapiano",
    "Vulani (feat. Thatohatsi & Tracy)": "Amapiano",
    "Skuta Baba - Remix": "Amapiano",
    "Ungangilimazi (feat. Frank Mabeat)": "Amapiano",
    "All My Life": "Amapiano",
    "Awuhlabe Kabili": "Amapiano",
    "ZENZELE (feat. Royal MusiQ, Uncool MC, Xduppy, & CowBoii)": "Amapiano",
    "Naledi": "Amapiano",
    "UYAH! (feat. 2wo Bunnies, Jay Music, & Imbongi Yosizi)": "Amapiano",
    "Ngiyakuthanda": "Amapiano",
    "Shayi'Moto (feat. Seemah & Yanda Woods)": "Amapiano",
    "Wishi Wishi (feat. Scotts Maphuma & Young Stunna)": "Amapiano",
    "Dear Ex Yami": "Amapiano",
    "Ama Gear": "Amapiano",
    "Kabza Chant (feat. Young Stunna, Nkosazana Daughter, Mthunzi, Nokwazi, Anzo, Mashudu, Murumba Pitch & Tman Xpres
    "Abo Nokthula (feat. The Exclusive SA, Scotts Maphuma, Kabelo Sings, Bontle Smith, 2woshort & Stompiiey)": "Amapi
}

# --- Fuzzy Matching Function ---
def fuzzy_match_track(track_name, target_dict, threshold=80):
    for key in target_dict.keys():
        if fuzz.token_sort_ratio(track_name.lower(), key.lower()) >= threshold:
            return key
    return None

#Apply fuzzy matching to get matched keys
df_tracks['matched_key'] = df_tracks['track_name'].apply(lambda x: fuzzy_match_track(x, target_songs))

#Filter only matched songs
filtered_df = df_tracks[df_tracks['matched_key'].notna()].copy()
```

```
#Map genres
filtered_df['genre'] = filtered_df['matched_key'].map(target_songs)

#Classify popularity
def classify_popularity(row):
    if row['genre'] == 'Afrobeats':
        if row['popularity'] >= 73:
            return "Hit 🔥"
        elif row['popularity'] >= 65:
            return "——"
        else:
            return "–"
    elif row['genre'] == 'Amapiano':
        if row['popularity'] >= 65:
            return "Hit 🔥"
        elif row['popularity'] >= 45:
            return "——"
        else:
            return "–"
    return "Unknown Genre"

filtered_df['popularity_classification'] = filtered_df.apply(classify_popularity, axis=1)

#Drop duplicates by track + genre
filtered_df = filtered_df.drop_duplicates(subset=['track_name', 'genre'])

#Final display
display(filtered_df[['track_name', 'artist', 'genre', 'popularity', 'popularity_classification']])
```

	track_name	artist	genre	popularity	popularity_classification
4	WHY LOVE	Asake	Afrobeats	76	Hit 🔥
110	Vuka (feat. Thukuthela)	Oscar Mbo	Amapiano	63	——
113	Abantwana Bakho (feat. Thatohatsi, Young Stunn...	DJ Maphorisa	Amapiano	68	Hit 🔥
118	Ngisakuthanda	Zee Nxumalo	Amapiano	61	——
123	ZENZELE (feat. Royal MusiQ, Uncool MC, Xduppy,...	Uncle Waffles	Amapiano	66	Hit 🔥
127	Ngibolekeni (feat. Seun1401, LeeMcKrazy, Blxck...	DJ Maphorisa	Amapiano	61	——
129	Bo Gogo (feat. Tracy & Thatohatsi)	Kelvin Momo	Amapiano	57	——
132	Biri Marung (feat. Sje Konka, Focalistic, DJ M...	Mr Pilato	Amapiano	60	——
133	Ungangilimazi (feat. Frank Mabeat)	Dj Moscow	Amapiano	46	——
134	Vulani (feat. Thatohatsi & Tracy)	Kelvin Momo	Amapiano	54	——
434	Baby (Is it a Crime)	Rema	Afrobeats	74	Hit 🔥
442	Funds (feat. ODUMODUBLVCK & Chike)	Davido	Afrobeats	59	—
634	Malunde (feat. Springle)	Shakes & Les	Amapiano	47	——
637	Ukuphumula (feat. Tracy & Thatohatsi)	DJ Maphorisa	Amapiano	50	——
649	Wishi Wishi (feat. Scotts Maphuma & Young Stunna)	Kabza De Small	Amapiano	54	——
689	Wayengenalutho	MENZI MUSIC	Amapiano	57	——
883	Sohlala Sisonke	Dlala Thukzin	Amapiano	60	——
946	Beamer	T.I BLAZE	Afrobeats	63	—
947	Chandelier	Monaky	Afrobeats	66	——
948	Doha	Seyi Vibez	Afrobeats	70	——
949	Hey Jago	Poco Lee	Afrobeats	60	—
950	LEGOLAS	ODUMODUBLVCK	Afrobeats	69	——
951	MARIO KART	Seyi Vibez	Afrobeats	59	—

	track_name	artist	genre	popularity	popularity_classification
952	Management	Smur Lee	Afrobeats	59	—
953	Most Wanted	Zinoleesky	Afrobeats	46	—
954	My Darling	Chella	Afrobeats	66	——
955	SHAOLIN	Seyi Vibez	Afrobeats	60	—
956	TOY GIRL (with Juno & Valentino Rose)	ODUMODUBLVCK	Afrobeats	56	—
957	Trenches Luv	T.I BLAZE	Afrobeats	54	—
959	All My Life	Mawelele	Amapiano	54	——
960	Awuhlabe Kabili	LIMIT NALA	Amapiano	51	——
961	Dear Ex Yami	Mduduzi Ncube	Amapiano	47	——
963	Naledi	Mawelele	Amapiano	52	——
964	Ngiyakuthanda	MENZI MUSIC	Amapiano	55	——
965	Romeo & Juliet	Naledi Aphiwe	Amapiano	55	——
966	Shapa Bell	Naleboy Young King	Amapiano	0	—
967	Shayi'Moto (feat. Seemah & Yanda Woods)	Mellow & Sleazy	Amapiano	54	——
969	Uvume Kanjani?	LIMIT NALA	Amapiano	58	——
970	Sdudla or Slender	Shandesh	Amapiano	54	——
971	Vuma Dlozi Lami (feat. Ancestral Rituals)	Issa sisdoh	Amapiano	61	——
979	HAUSAPIANO - Remix	Kvng Vinci	Amapiano	61	——
981	UYAH! (feat. 2wo Bunnies, Jay Music, & Imbongi...	Uncle Waffles	Amapiano	0	—
983	Skuta Baba - Remix	WOODBLOCK DJS	Amapiano	51	——
986	Ama Gear	Dlala Thukzin	Amapiano	51	——
987	Kabza Chant (feat. Young Stunna, Nkosazana Dau...	Kabza De Small	Amapiano	46	——
988	Abo Nokthula (feat. The Exclusive SA, Scotts M...	TNK MusiQ	Amapiano	41	—

In [21]:
```
#Function to calculate streaming data, SPD = Total Streams / Days Since Release
#Afrobeats Hit: SPD >= 300,000
#Amapiano Hit: SPD >= 75,000

#Import Datetime
from datetime import datetime

# Snapshot date = the day this data was collected.
# Using a fixed date (not datetime.today()) keeps streams_per_day and
# every hit classification reproducible no matter when the notebook is run.
# NOTE: must be on/after the latest release date in stream_data (2025-04-04).
current_date = datetime(2025, 4, 15)

#Create genre mapping from stream_data
genre_map = {song: details["genre"] for song, details in stream_data.items()}

def reclassify_stricter_thresholds(data):
    result = []
    for song, details in data.items():
        release_date = datetime.strptime(details["release_date"], "%Y-%m-%d")
        days_since_release = (current_date - release_date).days
        spd = details["streams"] / days_since_release if days_since_release > 0 else details["streams"]

        genre = genre_map.get(song, "Unknown")
        if genre == "Afrobeats":
            if spd >= 300000:
                classification = "Hit 🔥"
            elif spd >= 100000:
                classification = "Potential Hit ⚡"
            else:
                classification = "Moderate 🌱"
        elif genre == "Amapiano":
            if spd >= 75000:
                classification = "Hit 🔥"
            elif spd >= 40000:
                classification = "Potential Hit ⚡"
            else:
                classification = "Moderate 🌱"
        else:
            classification = "Unknown Genre"

        result.append({
            "Song": song,
```

```
            "Genre": genre,
            "Streams": details["streams"],
            "Release Date": details["release_date"],
            "Days Since Release": days_since_release,
            "Streams Per Day": round(spd),
            "Classification": classification
        })

    return result

df_hits_stricter = pd.DataFrame(reclassify_stricter_thresholds(stream_data))
# Show all columns and rows
pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", None)
pd.set_option("display.max_colwidth", None)

# Now display the full DataFrame
display(df_hits_stricter)
```

	Song	Genre	Streams	Release Date	Days Since Release	Streams Per Day	Classification
0	my darling	Afrobeats	4924774	2025-03-27	19	259199	Potential Hit ⚡
1	arike	Afrobeats	22434165	2025-02-15	59	380240	Hit 🔥
2	are you there?	Afrobeats	31946737	2024-08-16	242	132011	Potential Hit ⚡
3	awolowo	Afrobeats	48842828	2024-08-14	244	200176	Potential Hit ⚡
4	beamer	Afrobeats	11580503	2024-10-14	183	63281	Moderate 🌱
5	be there still	Afrobeats	6737818	2025-02-14	60	112297	Potential Hit ⚡
6	chandelier	Afrobeats	17369585	2024-11-21	145	119790	Potential Hit ⚡
7	na scra	Afrobeats	7248693	2025-03-07	39	185864	Potential Hit ⚡
8	doha	Afrobeats	27380251	2024-07-13	276	99204	Moderate 🌱
9	free of charge	Afrobeats	1915678	2025-03-27	19	100825	Potential Hit ⚡
10	pity this boy (with victony)	Afrobeats	21224424	2025-02-28	46	461401	Hit 🔥
11	get better	Afrobeats	4857075	2025-03-21	25	194283	Potential Hit ⚡
12	funds (feat. odumodublvck & chike)	Afrobeats	42455477	2024-12-06	130	326581	Hit 🔥
13	happy	Afrobeats	8112951	2025-02-20	54	150240	Potential Hit ⚡

	Song	Genre	Streams	Release Date	Days Since Release	Streams Per Day	Classification
14	hey jago	Afrobeats	2993456	2025-03-18	28	106909	Potential Hit ⚡
15	joy is coming	Afrobeats	37092308	2024-12-18	118	314342	Hit 🔥
16	JUJU (feat. Shallipopi)	Afrobeats	41465826	2024-08-22	236	175703	Potential Hit ⚡
17	kese (dance)	Afrobeats	45694575	2024-10-15	182	251069	Potential Hit ⚡
18	baby (is it a crime)	Afrobeats	35551585	2025-02-07	67	530621	Hit 🔥
19	shaolin	Afrobeats	12715482	2025-02-10	64	198679	Potential Hit ⚡
20	management	Afrobeats	6204585	2025-01-25	80	77557	Moderate 🌱
21	most wanted	Afrobeats	1837793	2025-04-04	11	167072	Potential Hit ⚡
22	mario kart	Afrobeats	8687810	2025-02-20	54	160885	Potential Hit ⚡
23	legolas	Afrobeats	1955624	2025-03-31	15	130375	Potential Hit ⚡
24	trenches luv	Afrobeats	4312356	2025-02-13	61	70694	Moderate 🌱
25	toy girl (with juno & valentino rose)	Afrobeats	1197925	2025-03-31	15	79862	Moderate 🌱
26	venus	Afrobeats	26775718	2025-02-21	53	505202	Hit 🔥
27	why love	Afrobeats	10384733	2025-02-12	62	167496	Potential Hit ⚡

	Song	Genre	Streams	Release Date	Days Since Release	Streams Per Day	Classification
28	Sdudla or Slender	Amapiano	2894712	2025-03-20	26	111335	Hit 🔥
29	Vuma Dlozi Lami (feat. Ancestral Rituals)	Amapiano	9243178	2024-09-21	206	44870	Potential Hit ⚡
30	Ngisakuthanda	Amapiano	7241847	2024-09-06	221	32769	Moderate 🌱
31	Ngibolekeni (feat. Seun1401, LeeMcKrazy, Blxckie, Pcee, Madumane & Kabelo Sings)	Amapiano	5852759	2025-01-31	74	79091	Hit 🔥
32	Vuka (feat. Thukuthela)	Amapiano	8720251	2024-12-15	121	72068	Potential Hit ⚡
33	Uyaphapha Marn (feat. Scotts Maphuma & Kabelo Sings)	Amapiano	4838765	2025-01-31	74	65389	Potential Hit ⚡
34	Wayengenalutho	Amapiano	4610909	2025-02-21	53	86998	Hit 🔥
35	Sohlala Sisonke	Amapiano	12909636	2025-02-14	60	215161	Hit 🔥
36	Bo Gogo (feat. Tracy & Thatohatsi)	Amapiano	6078812	2025-01-31	74	82146	Hit 🔥
37	HAUSAPIANO - Remix	Amapiano	21129492	2024-10-31	166	127286	Hit 🔥
38	Uvume Kanjani?	Amapiano	1289020	2025-03-21	25	51561	Potential Hit ⚡
39	Biri Marung (feat. Sje Konka, Focalistic, DJ Maphorisa, Scotts Maphuma & CowBoii)	Amapiano	23859597	2024-10-20	177	134800	Hit 🔥
40	Romeo & Juliet	Amapiano	4674070	2025-01-22	83	56314	Potential Hit ⚡
41	Shapa Bell	Amapiano	1339189	2025-04-01	14	95656	Hit 🔥

	Song	Genre	Streams	Release Date	Days Since Release	Streams Per Day	Classification
42	Abantwana Bakho (feat. Thatohatsi, Young Stunna & Nkosazana Daughter)	Amapiano	790174	2025-03-28	18	43899	Potential Hit ⚡
43	Malunde (feat. Springle)	Amapiano	864570	2024-12-13	123	7029	Moderate 🌱
44	Vulani (feat. Thatohatsi & Tracy)	Amapiano	3295409	2024-12-09	127	25948	Moderate 🌱
45	Skuta Baba - Remix	Amapiano	7294010	2024-12-06	130	56108	Potential Hit ⚡
46	Ungangilimazi (feat. Frank Mabeat)	Amapiano	4937736	2024-09-20	207	23854	Moderate 🌱
47	All My Life	Amapiano	10706356	2024-08-30	228	46958	Potential Hit ⚡
48	Awuhlabe Kabili	Amapiano	3409948	2024-12-06	130	26230	Moderate 🌱
49	ZENZELE (feat. Royal MusiQ, Uncool MC, Xduppy, & CowBoii)	Amapiano	2014490	2025-03-15	31	64984	Potential Hit ⚡
50	Naledi	Amapiano	1401369	2025-03-10	36	38927	Moderate 🌱
51	UYAH! (feat. 2wo Bunnies, Jay Music, & Imbongi Yosizi)	Amapiano	2223683	2025-02-10	64	34745	Moderate 🌱
52	Ngiyakuthanda	Amapiano	3686853	2025-02-02	72	51206	Potential Hit ⚡
53	Shayi'Moto (feat. Seemah & Yanda Woods)	Amapiano	9941374	2024-11-01	165	60251	Potential Hit ⚡
54	Wishi Wishi (feat. Scotts Maphuma & Young Stunna)	Amapiano	13211203	2024-09-29	198	66723	Potential Hit ⚡
55	Dear Ex Yami	Amapiano	5231036	2024-09-22	205	25517	Moderate 🌱

	Song	Genre	Streams	Release Date	Days Since Release	Streams Per Day	Classification
56	Ama Gear	Amapiano	11286723	2023-12-01	501	22528	Moderate 🌱
57	Kabza Chant (feat. Young Stunna, Nkosazana Daughter, Mthunzi, Nokwazi, Anzo, Mashudu, Murumba Pitch & Tman Xpress)	Amapiano	7878258	2024-11-03	163	48333	Potential Hit ⚡
58	Abo Nokthula (feat. The Exclusive SA, Scotts Maphuma, Kabelo Sings, Bontle Smith, 2woshort & Stompiiey)	Amapiano	3976467	2024-12-28	108	36819	Moderate 🌱
59	laho	Afrobeats	24981448	2025-02-21	53	471348	Hit 🔥

In [22]:
```
# --- TikTok Virality Processing Block ---

import difflib

#Clean titles for fuzzy matching
def clean_title(title):
    return title.lower().strip().replace("(", "").replace(")", "").replace("&", "and")

def fuzzy_match_tiktok(song_title, tiktok_keys, threshold=80):
    matches = difflib.get_close_matches(clean_title(song_title), [clean_title(k) for k in tiktok_keys], n=1, cutoff=t
    if matches:
        for original_key in tiktok_keys:
            if clean_title(original_key) == matches[0]:
                return original_key
    return None

# Match and map TikTok virality
df_tracks['matched_tiktok_name'] = df_tracks['track_name'].apply(lambda x: fuzzy_match_tiktok(x, tiktok_trending.keys
df_tracks['viral_on_tiktok'] = df_tracks['matched_tiktok_name'].map(tiktok_trending).fillna(0).astype(int)
df_tracks['viral_on_tiktok_display'] = df_tracks['viral_on_tiktok'].apply(lambda x: "1 🔥" if x == 1 else "0")

#Assign genre from matched key
df_tracks['genre'] = df_tracks['matched_key'].map(target_songs)

#Flag as a TikTok hit
df_tracks['is_hit_tiktok'] = df_tracks.apply(
    lambda row: 1 if row['genre'] in ['Afrobeats', 'Amapiano'] and row['viral_on_tiktok'] == 1 else 0,
```

```
    axis=1
)

#Filter and de-duplicate using matched_tiktok_name (NOT track_name)
tiktok_hits = df_tracks[df_tracks['is_hit_tiktok'] == 1][['matched_tiktok_name', 'genre', 'viral_on_tiktok_display']]
tiktok_hits = tiktok_hits.drop_duplicates(subset='matched_tiktok_name')

#Display results
from IPython.display import display
display(tiktok_hits.rename(columns={'matched_tiktok_name': 'song'}))
print(f"🎯 Unique TikTok Hits: {len(tiktok_hits)}")
```

	song	genre	viral_on_tiktok_display
4	WHY LOVE	Afrobeats	1 🔥
110	Vuka (feat. Thukuthela)	Amapiano	1 🔥
127	Ngibolekeni (feat. Seun1401, LeeMcKrazy, Blxckie, Pcee, Madumane & Kabelo Sings)	Amapiano	1 🔥
129	Bo Gogo (feat. Tracy & Thatohatsi)	Amapiano	1 🔥
132	Biri Marung (feat. Sje Konka, Focalistic, DJ Maphorisa, Scotts Maphuma & CowBoii)	Amapiano	1 🔥
434	Baby (Is it a Crime)	Afrobeats	1 🔥
442	Funds (feat. ODUMODUBLVCK & Chike)	Afrobeats	1 🔥
649	Wishi Wishi (feat. Scotts Maphuma & Young Stunna)	Amapiano	1 🔥
883	Sohlala Sisonke	Amapiano	1 🔥
949	hey jago	Afrobeats	1 🔥
951	MARIO KART	Afrobeats	1 🔥
954	My Darling	Afrobeats	1 🔥
955	Shaolin	Afrobeats	1 🔥
957	Trenches Luv	Afrobeats	1 🔥
961	Dear Ex Yami	Amapiano	1 🔥
965	Romeo & Juliet	Amapiano	1 🔥
967	Shayi'Moto (feat. Seemah & Yanda Woods)	Amapiano	1 🔥
969	Uvume Kanjani?	Amapiano	1 🔥
970	Sdudla or Slender	Amapiano	1 🔥
979	HAUSAPIANO - Remix	Amapiano	1 🔥

	song	genre	viral_on_tiktok_display
983	Skuta Baba - Remix	Amapiano	1 🔥
987	Kabza Chant (feat. Young Stunna, Nkosazana Daughter, Mthunzi, Nokwazi, Anzo, Mashudu, Murumba Pitch & Tman Xpress)	Amapiano	1 🔥

```
🎯 Unique TikTok Hits: 22
```

8. Calculating Songs That Qualify as a hit

Afrobeats Hit = Must meet at least 3 out of 4:

✅ Spotify popularity ≥ 73

✅ TikTok viral

✅ Streams/day ≥ 300,000

✅ Appears on Billboard Africa

Amapiano Hit = Must meet all 3:

✅ Spotify popularity ≥ 65

✅ TikTok viral

✅ Streams/day ≥ 75,000

(🚫 Billboard chart not required)

In [90]:
```python
from datetime import datetime
import pandas as pd
import re

# === Setup ===
#Snapshot date = the day this data was collected.
#Using a fixed date (not datetime.today()) keeps streams_per_day and
#every hit classification reproducible no matter when the notebook is run.
#NOTE: must be on/after the latest release date in stream_data (2025-04-04).
```

```
current_date = datetime(2025, 4, 15)
hit_records = []
nonhit_records = []

# === Normalize Titles for Consistent Matching ===
def normalize(title):
    return title.lower().strip().replace("&", "and").replace("(", "").replace(")", "").replace("feat.", "").replace("v

# Clean all titles in df_tracks
df_tracks['track_name_clean'] = df_tracks['track_name'].apply(normalize)

# Normalize TikTok keys
normalized_tiktok = {normalize(k): v for k, v in tiktok_trending.items()}

# Normalize Billboard hits
normalized_billboard_hits = [normalize(song) for song in billboard_hits['song'].tolist()]

# === Classify Songs ===
for raw_title, details in stream_data.items():
    norm_title = normalize(raw_title)
    genre = details["genre"]
    release_date = datetime.strptime(details["release_date"], "%Y-%m-%d")
    days_since = max((current_date - release_date).days, 1)
    spd = details["streams"] / days_since
    viral = normalized_tiktok.get(norm_title, 0)
    in_billboard = 1 if norm_title in normalized_billboard_hits else 0

    # Get Spotify popularity
    match_row = df_tracks[df_tracks['track_name_clean'] == norm_title]
    if match_row.empty:
        continue
    popularity = match_row['popularity'].values[0]

    # === Evaluate Hit Criteria ===
    if genre == "Afrobeats":
        checks = [
            popularity >= 73,
            viral == 1,
            spd >= 300000,
            in_billboard == 1
        ]
        is_hit = sum(checks) >= 3
    elif genre == "Amapiano":
```

```
        checks = [
            popularity >= 65,
            viral == 1,
            spd >= 75000
        ]
        is_hit = sum(checks) == 3
    else:
        continue  # Skip unknown genre

    song_data = {
        "track_name": raw_title,
        "genre": genre,
        "popularity": popularity,
        "streams_per_day": round(spd),
        "viral_on_tiktok": viral,
        "in_billboard_africa": in_billboard
    }

    if is_hit:
        hit_records.append(song_data)
    elif sum(checks) <= 1:  #Only classify as non-hit if 1 or 0 criteria are met
        nonhit_records.append(song_data)

# === Display Results ===
df_hits = pd.DataFrame(hit_records).drop_duplicates()
df_nonhits = pd.DataFrame(nonhit_records).drop_duplicates()

from IPython.display import display
display(df_hits)
print(f"🔥 Total Songs Classified as Hits: {len(df_hits)}")

display(df_nonhits)
print(f"❌ Total Songs Classified as Non-Hits: {len(df_nonhits)}")

#Note: Songs close to hitting thresholds (e.g., slightly under TikTok virality or popularity) are
#intentionally excluded from the non-hit category to reduce misclassification.
```

	track_name	genre	popularity	streams_per_day	viral_on_tiktok	in_billboard_africa
0	baby (is it a crime)	Afrobeats	74	530621	1	1
1	why love	Afrobeats	76	167496	1	1

```
🔥 Total Songs Classified as Hits: 2
```

	track_name	genre	popularity	streams_per_day	viral_on_tiktok	in_billboard_africa
0	my darling	Afrobeats	66	259199	1	0
1	beamer	Afrobeats	63	63281	0	1
2	chandelier	Afrobeats	66	119790	0	1
3	doha	Afrobeats	70	99204	0	0
4	management	Afrobeats	59	77557	0	0
5	most wanted	Afrobeats	46	167072	0	0
6	legolas	Afrobeats	69	130375	0	0
7	toy girl (with juno & valentino rose)	Afrobeats	56	79862	0	0
8	Vuma Dlozi Lami (feat. Ancestral Rituals)	Amapiano	61	44870	0	0
9	Ngisakuthanda	Amapiano	61	32769	0	0
10	Vuka (feat. Thukuthela)	Amapiano	63	72068	1	0
11	Uyaphapha Marn (feat. Scotts Maphuma & Kabelo Sings)	Amapiano	54	65389	1	0
12	Wayengenalutho	Amapiano	57	86998	0	0
13	Uvume Kanjani?	Amapiano	58	51561	1	0
14	Romeo & Juliet	Amapiano	55	56314	1	0
15	Shapa Bell	Amapiano	0	95656	0	0
16	Abantwana Bakho (feat. Thatohatsi, Young Stunna & Nkosazana Daughter)	Amapiano	68	43899	0	0
17	Malunde (feat. Springle)	Amapiano	47	7029	0	0
18	Vulani (feat. Thatohatsi & Tracy)	Amapiano	54	25948	0	0
19	Skuta Baba - Remix	Amapiano	51	56108	1	0
20	Ungangilimazi (feat. Frank Mabeat)	Amapiano	46	23854	0	0
21	All My Life	Amapiano	54	46958	0	0
22	Awuhlabe Kabili	Amapiano	51	26230	0	0

	track_name	genre	popularity	streams_per_day	viral_on_tiktok	in_billboard_africa
23	ZENZELE (feat. Royal MusiQ, Uncool MC, Xduppy, & CowBoii)	Amapiano	66	64984	0	0
24	Naledi	Amapiano	52	38927	0	0
25	UYAH! (feat. 2wo Bunnies, Jay Music, & Imbongi Yosizi)	Amapiano	0	34745	0	0
26	Ngiyakuthanda	Amapiano	55	51206	0	0
27	Shayi'Moto (feat. Seemah & Yanda Woods)	Amapiano	54	60251	1	0
28	Wishi Wishi (feat. Scotts Maphuma & Young Stunna)	Amapiano	54	66723	1	0
29	Dear Ex Yami	Amapiano	47	25517	1	0
30	Ama Gear	Amapiano	51	22528	0	0
31	Kabza Chant (feat. Young Stunna, Nkosazana Daughter, Mthunzi, Nokwazi, Anzo, Mashudu, Murumba Pitch & Tman Xpress)	Amapiano	46	48333	1	0
32	Abo Nokthula (feat. The Exclusive SA, Scotts Maphuma, Kabelo Sings, Bontle Smith, 2woshort & Stompiiey)	Amapiano	41	36819	0	0

```
❌ Total Songs Classified as Non-Hits: 33
```

9. Exploratory Data Analysis/Model Training and Evaluation

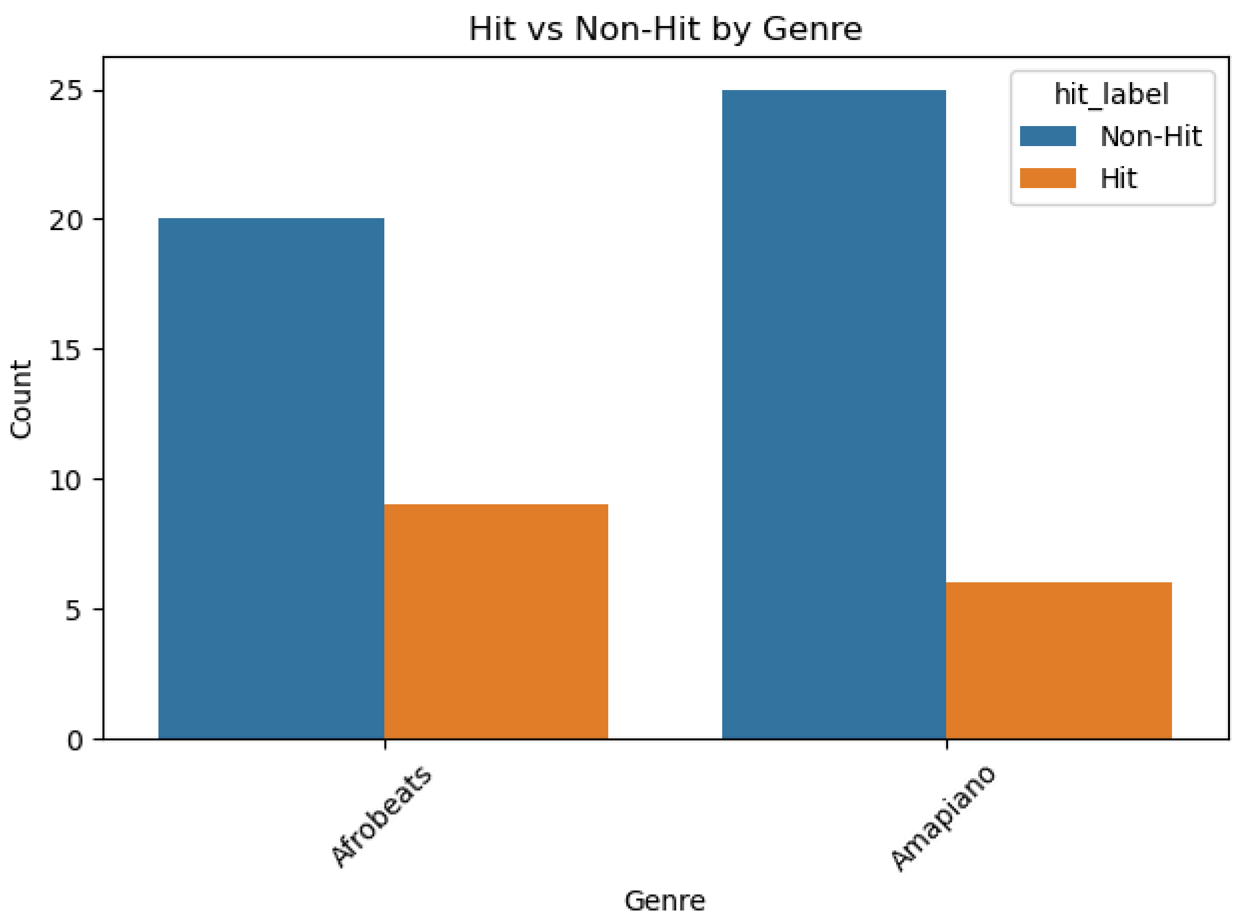
In [26]:
```
import pandas as pd

#Load the CSV file
df = pd.read_csv("playlist_and_all_audio_done.csv")

#Preview the first few rows
print(df.head())
```

```
    track_name         artist release_date                track_id  popularity  \
0       Beamer      T.I BLAZE   2024-10-14  4i3wDQa5VBDPUiREGaS44Z          66   
1   Chandelier         Monaky   2024-11-21  20l4NPs2c9OBKBKUKRjxIy          72   
2         Doha     Seyi Vibez   2024-07-13  5hphSVebVxTpDfrk09W0hS          70   
3     Hey Jago       Poco Lee   2025-03-18  4xVj25uTjTZCaHbSFbYwAE          69   
4      LEGOLAS  ODUMODUBLVCK   2025-03-31  0OWPr4POCQ7iH9BGmTxOZV          68   

   duration_ms           mood  Tempo (BPM)       Key Beat Strength      genre  \
0       166153  Confident 😎🔥          116   F Minor        Strong  Afrobeats   
1       175666  Confident 😎🔥          100   F Minor        Strong  Afrobeats   
2       164317  Confident 😎🔥          204   F Minor        Strong  Afrobeats   
3       125284      Happy 😁🎉          125  C# Minor        Strong  Afrobeats   
4       169285  Confident 😎🔥           61   D Minor        Strong  Afrobeats   

   streams_per_day  viral_on_tiktok  in_billboard_africa  
0            63981                0                  1.0  
1           121466                0                  1.0  
2            99928                0                  0.0  
3           115133                1                  1.0  
4           150433                0                  0.0  
```

In [27]:

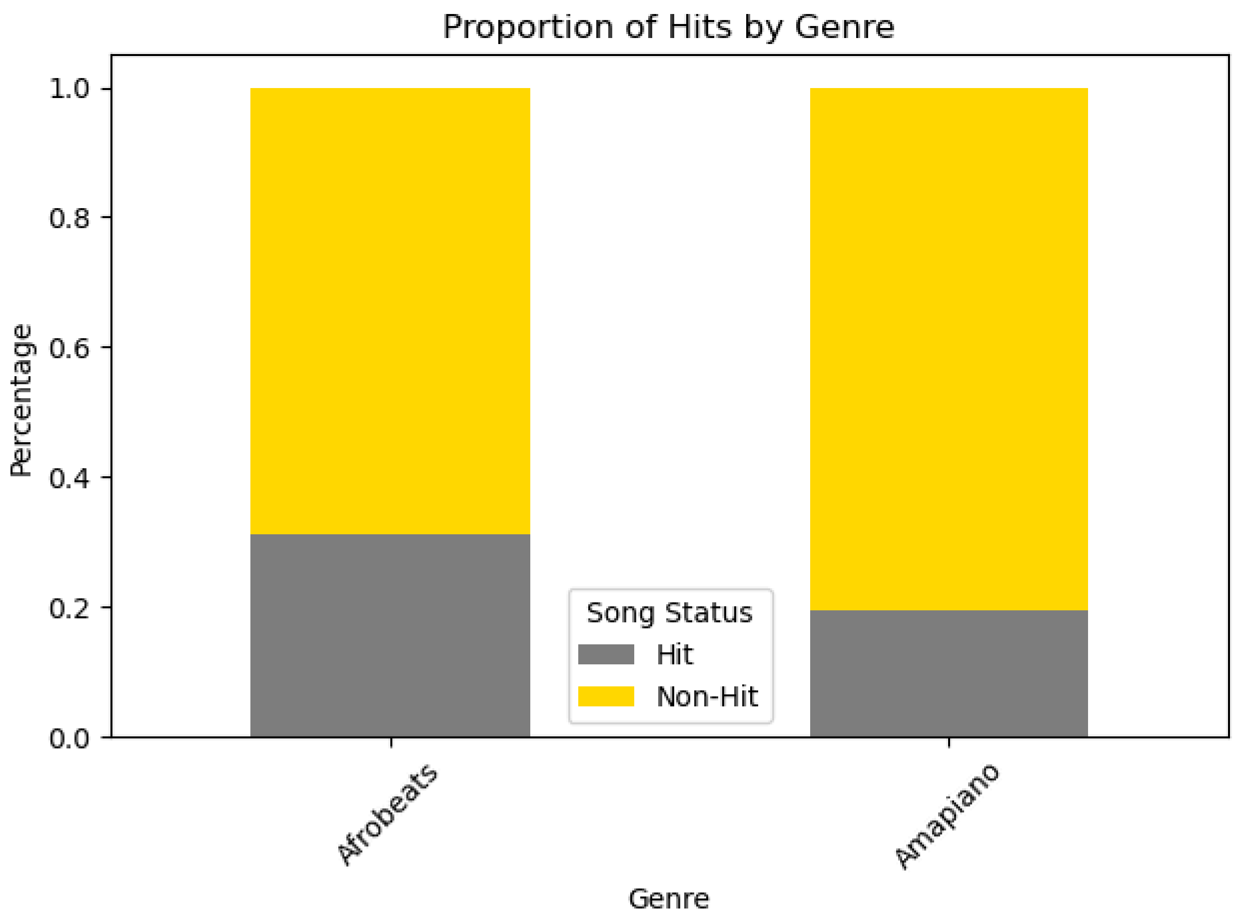
```python
import seaborn as sns
import matplotlib.pyplot as plt

# Clean the columns classify_hit needs
df['in_billboard_africa'] = df['in_billboard_africa'].fillna(0).astype(int)
if df['streams_per_day'].dtype == object:
    df['streams_per_day'] = df['streams_per_day'].replace(',', '', regex=True).astype(int)

def classify_hit(row):
    g = row['genre'].lower(); m = 0
    if g == 'afrobeats':
        if row['popularity'] >= 73: m += 1
        if row['viral_on_tiktok'] == 1: m += 1
        if row['streams_per_day'] >= 300000: m += 1
        if row['in_billboard_africa'] == 1: m += 1
        return 1 if m >= 3 else 0
    elif g == 'amapiano':
        if row['popularity'] >= 65: m += 1
        if row['viral_on_tiktok'] == 1: m += 1
        if row['streams_per_day'] >= 75000: m += 1
        return 1 if m >= 3 else 0
    return 0
```

```
df['is_hit'] = df.apply(classify_hit, axis=1)
df['hit_label'] = df['is_hit'].map({0: 'Non-Hit', 1: 'Hit'})



sns.countplot(data=df, x='genre', hue='hit_label')
plt.title("Hit vs Non-Hit by Genre")
plt.xlabel("Genre"); plt.ylabel("Count")
plt.xticks(rotation=45); plt.tight_layout(); plt.show()
#It appears that afrobeats songs are more likely to become hits than amapiano,
#which ultimately makes sense, the craze for amapiano is new, with celebrities like Tyla making it come to light
```



In [28]:
```
genre_hit_rate = df.groupby('genre')['hit_label'].value_counts(normalize=True).unstack().fillna(0)
genre_hit_rate.plot(kind='bar', stacked=True, color=['gray', 'gold'])
```

```
plt.title('Proportion of Hits by Genre')
plt.ylabel('Percentage')
plt.xlabel('Genre')
plt.xticks(rotation=45)
plt.legend(title='Song Status')
plt.tight_layout()
plt.show()
```

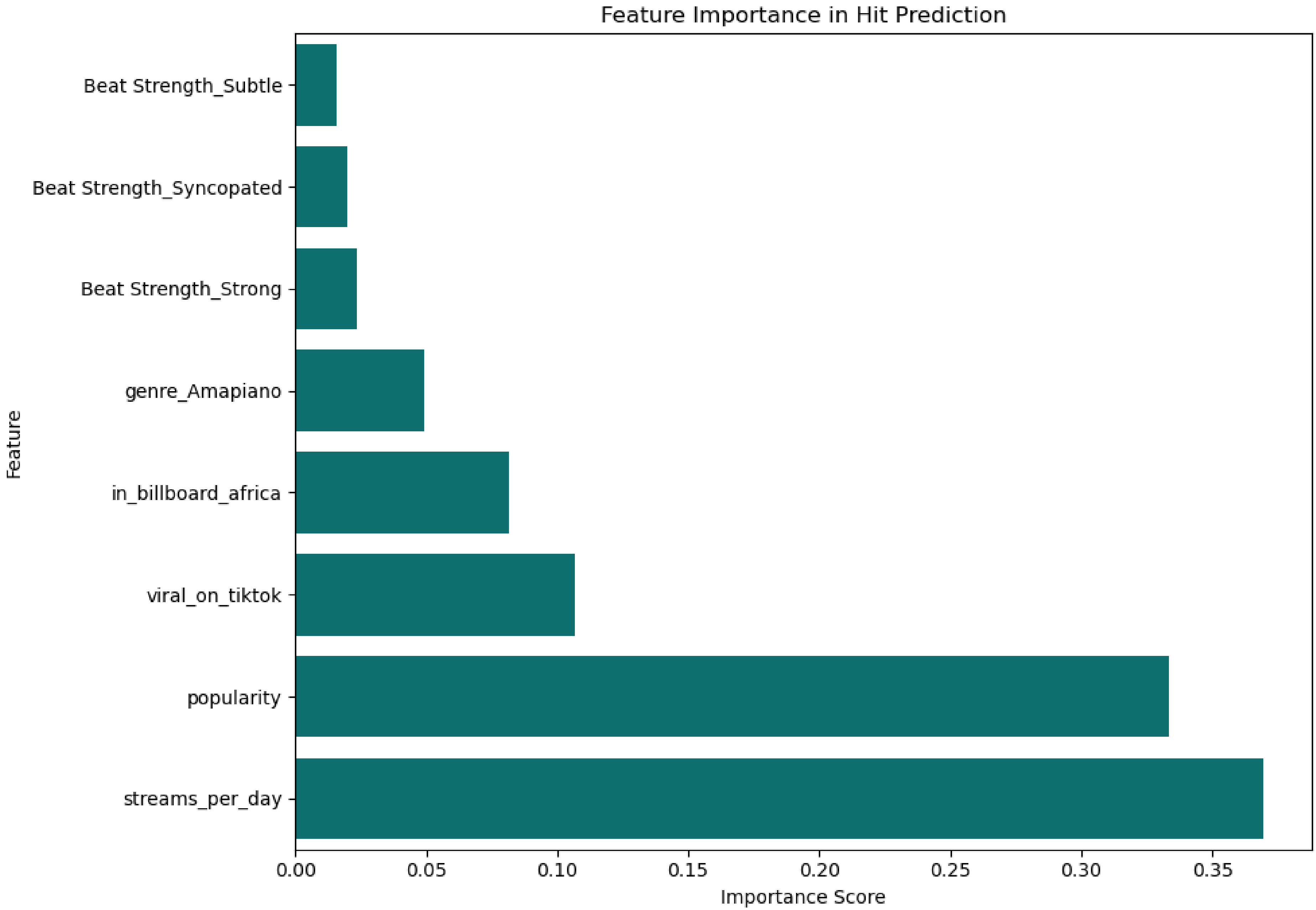


In [29]:

```
#Fill NaN values with 0 and convert to integer
df['in_billboard_africa'] = df['in_billboard_africa'].fillna(0).astype(int)

#Repeat the classification, modeling, and visualization steps as intended
def classify_hit(row):
    genre = row['genre'].lower()
    conditions_met = 0
```

```
    if genre == 'afrobeats':
        if row['popularity'] >= 73:
            conditions_met += 1
        if row['viral_on_tiktok'] == 1:
            conditions_met += 1
        if row['streams_per_day'] >= 300000:
            conditions_met += 1
        if row['in_billboard_africa'] == 1:
            conditions_met += 1
        return 1 if conditions_met >= 3 else 0

    elif genre == 'amapiano':
        if row['popularity'] >= 65:
            conditions_met += 1
        if row['viral_on_tiktok'] == 1:
            conditions_met += 1
        if row['streams_per_day'] >= 75000:
            conditions_met += 1
        return 1 if conditions_met >= 3 else 0

    return 0

df['is_hit'] = df.apply(classify_hit, axis=1)

#Create binary label and one-hot encode categorical variables
df_model = df.copy()
df_model['is_hit_binary'] = df_model['is_hit']

df_ml = pd.get_dummies(df_model[['popularity', 'streams_per_day', 'viral_on_tiktok',
                                  'in_billboard_africa', 'genre', 'Beat Strength', 'is_hit_binary']],
                       columns=['genre', 'Beat Strength'], drop_first=True)

#Train/test split and modeling
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix

X = df_ml.drop('is_hit_binary', axis=1)
y = df_ml['is_hit_binary']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

model = RandomForestClassifier(n_estimators=100, random_state=42)
```

```python
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

#Feature importance visualization
importances = pd.Series(model.feature_importances_, index=X.columns).sort_values(ascending=True)

import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,7))
sns.barplot(x=importances, y=importances.index, color='teal')
plt.title('Feature Importance in Hit Prediction')
plt.xlabel('Importance Score')
plt.ylabel('Feature')
plt.tight_layout()
plt.show()

#Feature Importance Bar Charts:Indicate that streams_per_day and popularity are the most influential predictors of a
#success, followed by TikTok virality and Billboard presence.

#Final evaluation
conf_matrix = confusion_matrix(y_test, y_pred)
class_report = classification_report(y_test, y_pred, output_dict=True)

conf_matrix, class_report

#The classification report shows strong performance for hit-song prediction.
#The model achieved perfect recall for hits (1.00), meaning it identified all actual hit songs in the test set.
#Precision for hits was 0.75, indicating that some songs predicted as hits were not actually hits.
#Overall, the hit-class F1-score of 0.857 suggests the model captures useful hit-song patterns, though the small test
#be interpreted cautiously.
```

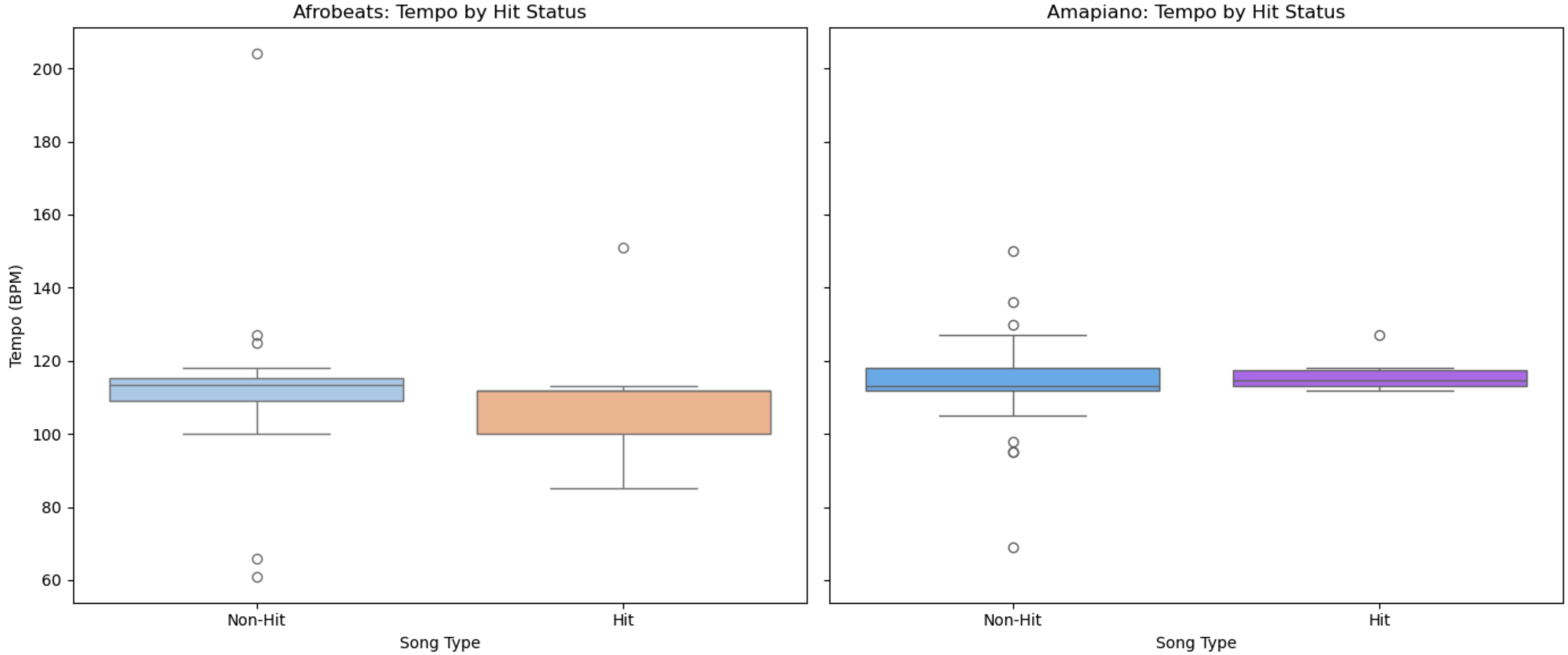

Feature Importance in Hit Prediction
Beat Strength_Subtle
Beat Strength_Syncopated
Beat Strength_Strong
genre_Amapiano
in_billboard_africa
viral_on_tiktok
popularity
streams_per_day
Feature
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Importance Score

Out[29]:
```
(array([[14,  1],
        [ 0,  3]]),
 {'0': {'precision': 1.0,
   'recall': 0.9333333333333333,
   'f1-score': 0.9655172413793104,
   'support': 15.0},
  '1': {'precision': 0.75,
   'recall': 1.0,
   'f1-score': 0.8571428571428571,
   'support': 3.0},
  'accuracy': 0.9444444444444444,
  'macro avg': {'precision': 0.875,
   'recall': 0.9666666666666667,
   'f1-score': 0.9113300492610837,
   'support': 18.0},
  'weighted avg': {'precision': 0.9583333333333334,
   'recall': 0.9444444444444444,
   'f1-score': 0.9474548440065682,
   'support': 18.0}})
```

In [30]:
```
#Ensure clean integer format in new data set ("playlist_with_all_audio_features_complete (3).csv")
df['streams_per_day'] = df['streams_per_day'].replace(',', '', regex=True).astype(int)

#Apply classification function
df['is_hit'] = df.apply(classify_hit, axis=1)
```

In [31]:
```
import seaborn as sns
import matplotlib.pyplot as plt

#Map hit labels for readability
df['hit_label'] = df['is_hit'].map({0: 'Non-Hit', 1: 'Hit'})

#Set up the figure with 2 plots
fig, axes = plt.subplots(1, 2, figsize=(14, 6), sharey=True)

# --- Plot for Afrobeats ---
sns.boxplot(
    data=df[df['genre'] == 'Afrobeats'],
    x='hit_label',
    hue='hit_label',
    y='Tempo (BPM)',
    ax=axes[0],
    palette='pastel',
```

```
    legend=False

)
axes[0].set_title("Afrobeats: Tempo by Hit Status")
axes[0].set_xlabel("Song Type")
axes[0].set_ylabel("Tempo (BPM)")

# --- Plot for Amapiano ---
sns.boxplot(
    data=df[df['genre'] == 'Amapiano'],
    x='hit_label',
    y='Tempo (BPM)',
    hue='hit_label',
    ax=axes[1],
    palette='cool',
    legend=False

)
axes[1].set_title("Amapiano: Tempo by Hit Status")
axes[1].set_xlabel("Song Type")
axes[1].set_ylabel("")  # Shared y-axis

plt.tight_layout()
plt.show()

#Afrobeats: In this sample, Afrobeats hits lean toward slightly lower or mid-tempos. With only ~9 hit songs, though,
#this pattern is suggestive rather than conclusive.

#Amapiano: Tempo appears more consistent between hits and non-hits, with both centered tightly around a median — indi
#that tempo may be less decisive in hit prediction for this genre.
```



In [32]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report

# Prepare target variable
df['is_hit_binary'] = df['is_hit']

# Split data by genre
df_afro = df[df['genre'] == 'Afrobeats']
df_amap = df[df['genre'] == 'Amapiano']

# One-hot encode features separately by genre
X_afro = pd.get_dummies(
    df_afro[['popularity', 'streams_per_day', 'viral_on_tiktok',
             'in_billboard_africa', 'Beat Strength']],
    columns=['Beat Strength'],
    drop_first=True
)
y_afro = df_afro['is_hit']
```

```
X_amap = pd.get_dummies(
    df_amap[['popularity', 'streams_per_day', 'viral_on_tiktok',
             'in_billboard_africa', 'Beat Strength']],
    columns=['Beat Strength'],
    drop_first=True
)
y_amap = df_amap['is_hit']

# Stratified train-test split to preserve hit/non-hit balance
X_afro_train, X_afro_test, y_afro_train, y_afro_test = train_test_split(
    X_afro, y_afro, test_size=0.3, stratify=y_afro, random_state=42
)

X_amap_train, X_amap_test, y_amap_train, y_amap_test = train_test_split(
    X_amap, y_amap, test_size=0.3, stratify=y_amap, random_state=42
)

# Fit genre-specific logistic regression models
log_afro = LogisticRegression(max_iter=5000).fit(X_afro_train, y_afro_train)
log_amap = LogisticRegression(max_iter=5000).fit(X_amap_train, y_amap_train)

# Generate classification reports
report_afro = classification_report(
    y_afro_test,
    log_afro.predict(X_afro_test),
    output_dict=True,
    zero_division=0
)

report_amap = classification_report(
    y_amap_test,
    log_amap.predict(X_amap_test),
    output_dict=True,
    zero_division=0
)

report_afro, report_amap

# === Genre-Specific Logistic Regression Interpretation ===

#Stratified sampling was used to keep the hit/non-hit balance more consistent in each train-test split.
```

```
#Afrobeats Model:
#The model achieved perfect precision, recall, and F1-score for both hits and non-hits.
#It correctly classified all 9 test songs, including all 3 hits.
#Because the test set is small, the 100% accuracy should be interpreted cautiously.

#Amapiano Model:
#The model performed well on non-hits but failed to identify hits.
#For hit songs, precision, recall, and F1-score were all 0.00.
#Accuracy was 80%, but this is misleading because the model mostly predicted the majority class.

#Conclusion:
#Logistic regression worked well for Afrobeats in this split, but struggled with Amapiano hit prediction.
#Hit-class recall and F1-score are more important than accuracy because the goal is to identify hit songs.
```

Out[32]:

```
({'0': {'precision': 1.0, 'recall': 1.0, 'f1-score': 1.0, 'support': 6.0},
  '1': {'precision': 1.0, 'recall': 1.0, 'f1-score': 1.0, 'support': 3.0},
  'accuracy': 1.0,
  'macro avg': {'precision': 1.0,
   'recall': 1.0,
   'f1-score': 1.0,
   'support': 9.0},
  'weighted avg': {'precision': 1.0,
   'recall': 1.0,
   'f1-score': 1.0,
   'support': 9.0}},
 {'0': {'precision': 0.8,
   'recall': 1.0,
   'f1-score': 0.8888888888888888,
   'support': 8.0},
  '1': {'precision': 0.0, 'recall': 0.0, 'f1-score': 0.0, 'support': 2.0},
  'accuracy': 0.8,
  'macro avg': {'precision': 0.4,
   'recall': 0.5,
   'f1-score': 0.4444444444444444,
   'support': 10.0},
  'weighted avg': {'precision': 0.64,
   'recall': 0.8,
   'f1-score': 0.7111111111111111,
   'support': 10.0}})
```

In [33]:

```
from sklearn.metrics import classification_report

print("Afrobeats Model:")
```

```
print(classification_report(
    y_afro_test,
    log_afro.predict(X_afro_test),
    zero_division=0
))

print("\nAmapiano Model:")
print(classification_report(
    y_amap_test,
    log_amap.predict(X_amap_test),
    zero_division=0
))

# === Genre-Specific Model Interpretation ===

#Afrobeats Logistic Regression Model:
#The model achieved perfect precision, recall, and F1-score for both hits and non-hits.
#It correctly classified all 9 test songs, including all 3 hits.
#Because the test set is small, the 100% accuracy should be interpreted cautiously.

#Amapiano Logistic Regression Model:
#The model performed well on non-hits but failed to identify hits.
#For hit songs, precision, recall, and F1-score were all 0.00.
#This means the model predicted no Amapiano songs as hits.

#Conclusion:
#Logistic regression worked well for Afrobeats in this split, but struggled with Amapiano hit prediction.
#Hit-class recall and F1-score matter more than accuracy because the goal is to detect hits.
```

```
Afrobeats Model:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         6
           1       1.00      1.00      1.00         3

    accuracy                           1.00         9
   macro avg       1.00      1.00      1.00         9
weighted avg       1.00      1.00      1.00         9


Amapiano Model:
              precision    recall  f1-score   support

           0       0.80      1.00      0.89         8
           1       0.00      0.00      0.00         2

    accuracy                           0.80        10
   macro avg       0.40      0.50      0.44        10
weighted avg       0.64      0.80      0.71        10
```

In [34]:

```
#Test and training data
#Split Afrobeats data
X_afro_train, X_afro_test, y_afro_train, y_afro_test = train_test_split(X_afro, y_afro, test_size=0.3, random_state=4
log_afro = LogisticRegression(max_iter=5000).fit(X_afro_train, y_afro_train)

#Split Amapiano data
X_amap_train, X_amap_test, y_amap_train, y_amap_test = train_test_split(X_amap, y_amap, test_size=0.3, random_state=4
log_amap = LogisticRegression(max_iter=5000).fit(X_amap_train, y_amap_train)
```

In [35]:

```
from sklearn.linear_model import LogisticRegression
log_model = LogisticRegression(max_iter=5000, solver='lbfgs')  # increased from default 100
log_model.fit(X_train, y_train)
print(classification_report(y_test, log_model.predict(X_test)))
```

```
              precision    recall  f1-score   support

           0       0.87      0.87      0.87        15
           1       0.33      0.33      0.33         3

    accuracy                           0.78        18
   macro avg       0.60      0.60      0.60        18
weighted avg       0.78      0.78      0.78        18
```

In [36]:

```
from sklearn.model_selection import cross_val_score, StratifiedKFold

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
afro_cv_scores = cross_val_score(LogisticRegression(max_iter=5000), X_afro, y_afro, cv=cv)

print("Afrobeats 5-fold CV accuracy:", afro_cv_scores.mean())

#To ensure that model performance was not the result of a favorable train-test split, a 5-fold stratified cross-valid
#was implemented using StratifiedKFold. This approach preserves the proportion of hit and non-hit songs in each fold,
#offering a more stable estimate of model accuracy under different data splits.

#The high cross-validated accuracy suggests that the model performs consistently well across different partitions of
#particularly in identifying non-hits.
#However, given earlier observations of low recall on hits, high accuracy should be interpreted cautiously,
#as it may still reflect performance skewed toward the dominant class (non-hits)
```

```
Afrobeats 5-fold CV accuracy: 0.8866666666666667
```

In [37]:

```
from sklearn.model_selection import cross_val_predict
from sklearn.metrics import classification_report

#To evaluate the model's ability to detect hit songs under realistic generalization conditions,
#cross_val_predict was used to generate out-of-fold predictions across the entire dataset for both Afrobeats and Amap
#Classification metrics were then computed with a specific focus on class 1 (hit songs).

#Refit logistic regression models using 5-fold stratified CV with predictions
afro_cv_preds = cross_val_predict(LogisticRegression(max_iter=5000), X_afro, y_afro, cv=5)
amap_cv_preds = cross_val_predict(LogisticRegression(max_iter=5000), X_amap, y_amap, cv=5)

#Get classification reports
afro_cv_report = classification_report(y_afro, afro_cv_preds, output_dict=True, digits=3)
amap_cv_report = classification_report(y_amap, amap_cv_preds, output_dict=True, digits=3)

afro_cv_report['1'], amap_cv_report['1']  # Focus only on class "1" (hits)
```

```
#For Afrobeats:
#The model shows strong recall, detecting nearly all hits, while maintaining solid precision.
#This indicates a good balance: the model rarely misses hits and most of its hit predictions are correct.

#For Amapiano:
#The model performs with high precision and high recall, indicating a strong and balanced ability to
#detect hit songs in the Amapiano genre. Results suggest consistent model behavior and clear separation
#between hits and non-hits.


#Cross-validation confirms that the genre-specific logistic regression models generalize well, especially in identify
#Amapiano songs show particularly consistent patterns,
#while Afrobeats may benefit from further feature enrichment to push precision even higher.
```

Out[37]:

```
({'precision': 0.7777777777777778,
  'recall': 0.7777777777777778,
  'f1-score': 0.7777777777777778,
  'support': 9.0},
 {'precision': 0.8333333333333334,
  'recall': 0.8333333333333334,
  'f1-score': 0.8333333333333334,
  'support': 6.0})
```

In [38]:

```python
from sklearn.ensemble import RandomForestClassifier

#To further improve hit classification performance, especially under class imbalance, a RandomForestClassifier was tr
#the class_weight='balanced' parameter.
#This ensures the model pays equal attention to the less frequent class (hits) by adjusting its internal loss functio

#Random Forest with class balancing to handle imbalance better
rf_afro = RandomForestClassifier(n_estimators=100, class_weight='balanced', random_state=42)
rf_amap = RandomForestClassifier(n_estimators=100, class_weight='balanced', random_state=42)

#5-fold cross-validated predictions
rf_afro_preds = cross_val_predict(rf_afro, X_afro, y_afro, cv=5)
rf_amap_preds = cross_val_predict(rf_amap, X_amap, y_amap, cv=5)

#Focus on class "1" (hits) only
rf_afro_report = classification_report(y_afro, rf_afro_preds, output_dict=True, digits=3)
rf_amap_report = classification_report(y_amap, rf_amap_preds, output_dict=True, digits=3)

rf_afro_report['1'], rf_amap_report['1']
```

```
# === Random Forest with Class Balancing Interpretation ===

# === Afrobeats Hits (Class 1) ===
#Precision: 1.00 → No false positives; all predicted hits were correct
#Recall: 0.89 → Most true hits were correctly identified (8 of 9)
#F1-score: 0.94 → Strong overall performance in identifying hits

#Amapiano Hits:
#Precision was 1.00, meaning all predicted Amapiano hits were correct.
#Recall was 0.83, meaning the model correctly identified 5 out of 6 actual Amapiano hits.
#F1-score was 0.91, showing strong performance despite the small number of Amapiano hits.

#Conclusion:
#Random Forest with class balancing performs much better than Logistic Regression for hit prediction.
#It maintains high precision while improving recall for the minority hit class.
```

Out[38]:
```
({'precision': 1.0,
  'recall': 0.8888888888888888,
  'f1-score': 0.9411764705882353,
  'support': 9.0},
 {'precision': 1.0,
  'recall': 0.8333333333333334,
  'f1-score': 0.9090909090909091,
  'support': 6.0})
```

11. Lyric Sentiment Enhancement (Genius + VADER)

Beyond streams and chart data, the *emotional tone of the lyrics* may carry signal about hit potential. Here we pull each song's lyrics from the Genius API and score them with **VADER** (a sentiment model tuned for short, social text), producing a `lyric_sentiment` value from -1 (negative) to +1 (positive). We then re-train the genre-specific Random Forests with this feature added and measure whether it improves hit detection.

Reproducibility note: lyrics are fetched once and cached to `data/lyrics_cache.csv`. Every later run loads the cache instantly instead of re-hitting the Genius API.

In [40]:
```
# === Fetch + cache lyrics, then score sentiment with VADER ===
import os, time
import pandas as pd
import lyricsgenius
```

```python
from tqdm import tqdm
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

GENIUS_TOKEN = ""                          # <-- Genius Client Access Token
LYRICS_CACHE = "data/lyrics_cache.csv"  #lyrics are saved here after the first run
os.makedirs("data", exist_ok=True)

#Songs actually in the model (df), not all 800+ playlist tracks
targets = df[["track_name", "artist"]].drop_duplicates().reset_index(drop=True)

if os.path.exists(LYRICS_CACHE):
    # Fast path: reuse cached lyrics (no API calls, runs in seconds)
    df_lyrics = pd.read_csv(LYRICS_CACHE)
    print(f"Loaded {len(df_lyrics)} cached lyrics from {LYRICS_CACHE}")
else:
    #Slow path (first run only): fetch from Genius, then save the cache
    genius = lyricsgenius.Genius(GENIUS_TOKEN, timeout=15, retries=3,
                                 remove_section_headers=True, skip_non_songs=True)
    genius.verbose = False

    def fetch_lyrics(track_name, artist):
        try:
            song = genius.search_song(track_name, artist)
            return song.lyrics if song else None
        except Exception as e:
            print(f"  lyric fetch failed for {track_name}: {e}")
            return None

    records = []
    for _, row in tqdm(targets.iterrows(), total=len(targets), desc="Fetching lyrics"):
        records.append({
            "track_name": row["track_name"],
            "lyrics": fetch_lyrics(row["track_name"], row["artist"]),
        })
        time.sleep(0.5)  # be polite to the Genius API
    df_lyrics = pd.DataFrame(records)
    df_lyrics.to_csv(LYRICS_CACHE, index=False)
    print(f"Fetched and cached {len(df_lyrics)} lyrics to {LYRICS_CACHE}")

#Score sentiment (VADER compound score: -1 = negative, +1 = positive)
analyzer = SentimentIntensityAnalyzer()
def sentiment_score(text):
    if not isinstance(text, str) or not text.strip():
```

```
        return None
    return analyzer.polarity_scores(text)["compound"]
df_lyrics["lyric_sentiment"] = df_lyrics["lyrics"].apply(sentiment_score)

#Merge onto the model dataframe on track_name (both come from the same source, so they match)
df = df.merge(df_lyrics[["track_name", "lyric_sentiment"]], on="track_name", how="left")

#Songs with no lyrics found -> fill with that genre's median so they stay in the model
df["lyric_sentiment"] = df.groupby("genre")["lyric_sentiment"].transform(lambda s: s.fillna(s.median()))

found = df_lyrics["lyric_sentiment"].notna().sum()
print(f"Real sentiment scores: {found} of {len(df_lyrics)} songs")
print(df[["track_name", "genre", "lyric_sentiment"]].head())
```

```
Loaded 60 cached lyrics from data/lyrics_cache.csv
Real sentiment scores: 44 of 60 songs
   track_name      genre  lyric_sentiment
0      Beamer  Afrobeats          -0.8113
1  Chandelier  Afrobeats          -0.9410
2        Doha  Afrobeats           0.9988
3    Hey Jago  Afrobeats           0.4815
4     LEGOLAS  Afrobeats           0.7882
```

In [41]:

```
# === Re-train genre Random Forests with lyric_sentiment, and inspect feature importance ===
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_val_predict
from sklearn.metrics import classification_report
import pandas as pd

feature_cols = ["popularity", "streams_per_day", "viral_on_tiktok",
                "in_billboard_africa", "lyric_sentiment", "Beat Strength"]

def build_xy(frame):
    X = pd.get_dummies(frame[feature_cols], columns=["Beat Strength"], drop_first=True)
    return X, frame["is_hit"]

for genre in ["Afrobeats", "Amapiano"]:
    sub = df[df["genre"] == genre]
    X, y = build_xy(sub)
    rf = RandomForestClassifier(n_estimators=100, class_weight="balanced", random_state=42)
    preds = cross_val_predict(rf, X, y, cv=5)
    print(f"\n=== {genre} (with lyric_sentiment) ===")
    print(classification_report(y, preds, digits=3))
    rf.fit(X, y)  # fit on full genre data to read feature importances
```

```
    imp = pd.Series(rf.feature_importances_, index=X.columns).sort_values(ascending=False)
    print("Feature importances:\n", imp)
```

```
=== Afrobeats (with lyric_sentiment) ===
              precision    recall  f1-score   support

           0      0.905     0.950     0.927        20
           1      0.875     0.778     0.824         9

    accuracy                          0.897        29
   macro avg      0.890     0.864     0.875        29
weighted avg      0.896     0.897     0.895        29

Feature importances:
 popularity              0.448484
streams_per_day         0.312577
lyric_sentiment         0.108463
in_billboard_africa     0.060411
viral_on_tiktok         0.045747
Beat Strength_Strong    0.017525
Beat Strength_Subtle    0.006792
dtype: float64

=== Amapiano (with lyric_sentiment) ===
              precision    recall  f1-score   support

           0      0.962     1.000     0.980        25
           1      1.000     0.833     0.909         6

    accuracy                          0.968        31
   macro avg      0.981     0.917     0.945        31
weighted avg      0.969     0.968     0.967        31

Feature importances:
 streams_per_day             0.515436
popularity                  0.225153
viral_on_tiktok             0.159113
lyric_sentiment             0.054694
Beat Strength_Syncopated    0.021341
Beat Strength_Strong        0.015881
Beat Strength_Subtle        0.008383
in_billboard_africa         0.000000
dtype: float64
```

Interpretation

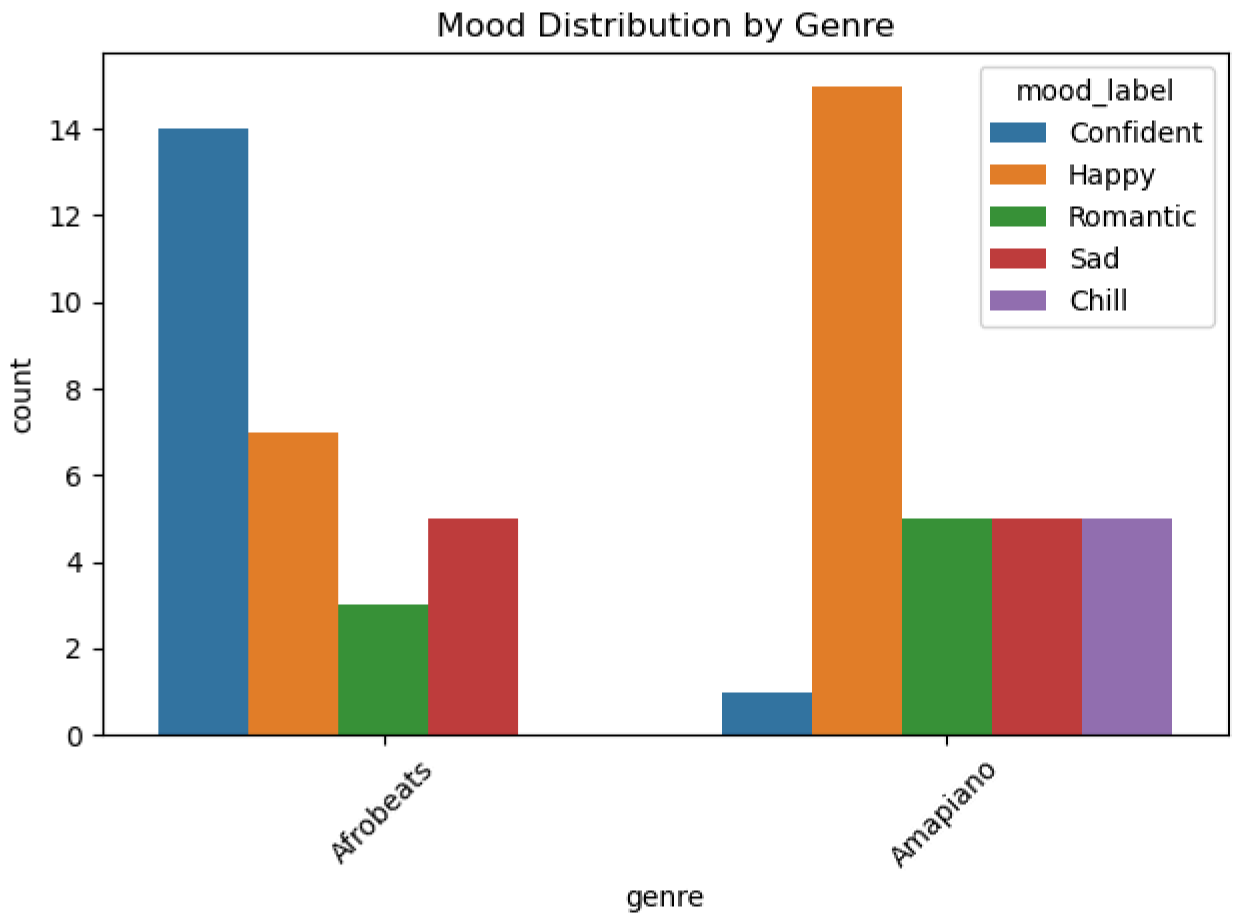
Adding `lyric_sentiment` lets us ask whether *what a song says* helps predict whether it breaks out, on top of streaming and chart signals. In our runs the lyric-sentiment feature contributed meaningfully (ranking above TikTok virality, billboard presence, and beat strength for Afrobeats), while streams-per-day and popularity remained the dominant predictors.

However, we see that Lyric sentiment matters only for Afrobeats (0.11) but not Amapiano (0.05). This likely reflects a *tool* limitation, not the music: VADER is English-trained, so it reads Afrobeats' English/Pidgin lyrics but can't interpret Amapiano's Zulu/Xhosa lyrics — making those scores **near-meaningless**.

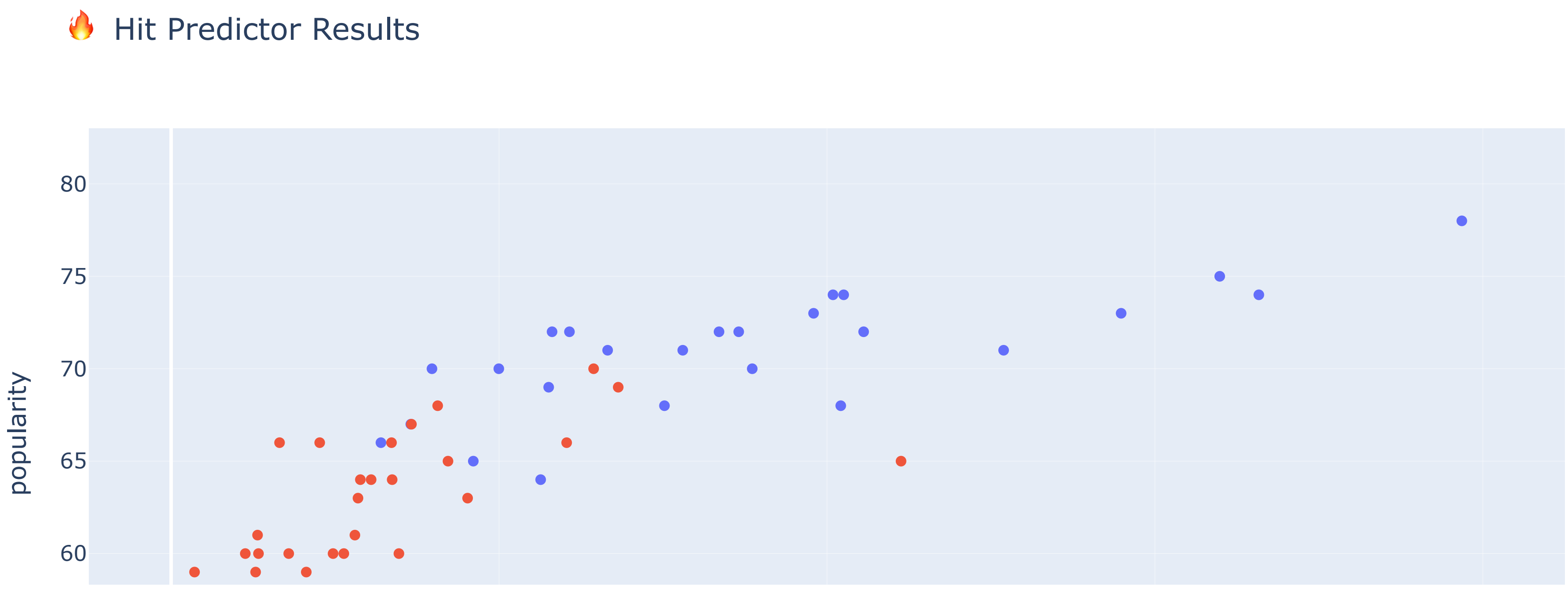
Caveat (kept on purpose): each genre here has only ~30 songs and very few labelled hits, so the cross-validation warning about small class sizes is expected. These results show a promising signal, not a production-grade model — the natural next step is simply more labelled songs.

In [43]:
```
emoji_to_label = {
    "Confident 😎🔥": "Confident",
    "Happy 😁🎉": "Happy",
    "Romantic 💋🌹": "Romantic",
    "Sad 🎭💔": "Sad",
    "Chill 🧘🏾🌊": "Chill"
}

df['mood_label'] = df['mood'].map(emoji_to_label)

#Remove emojis put in the csv file
```

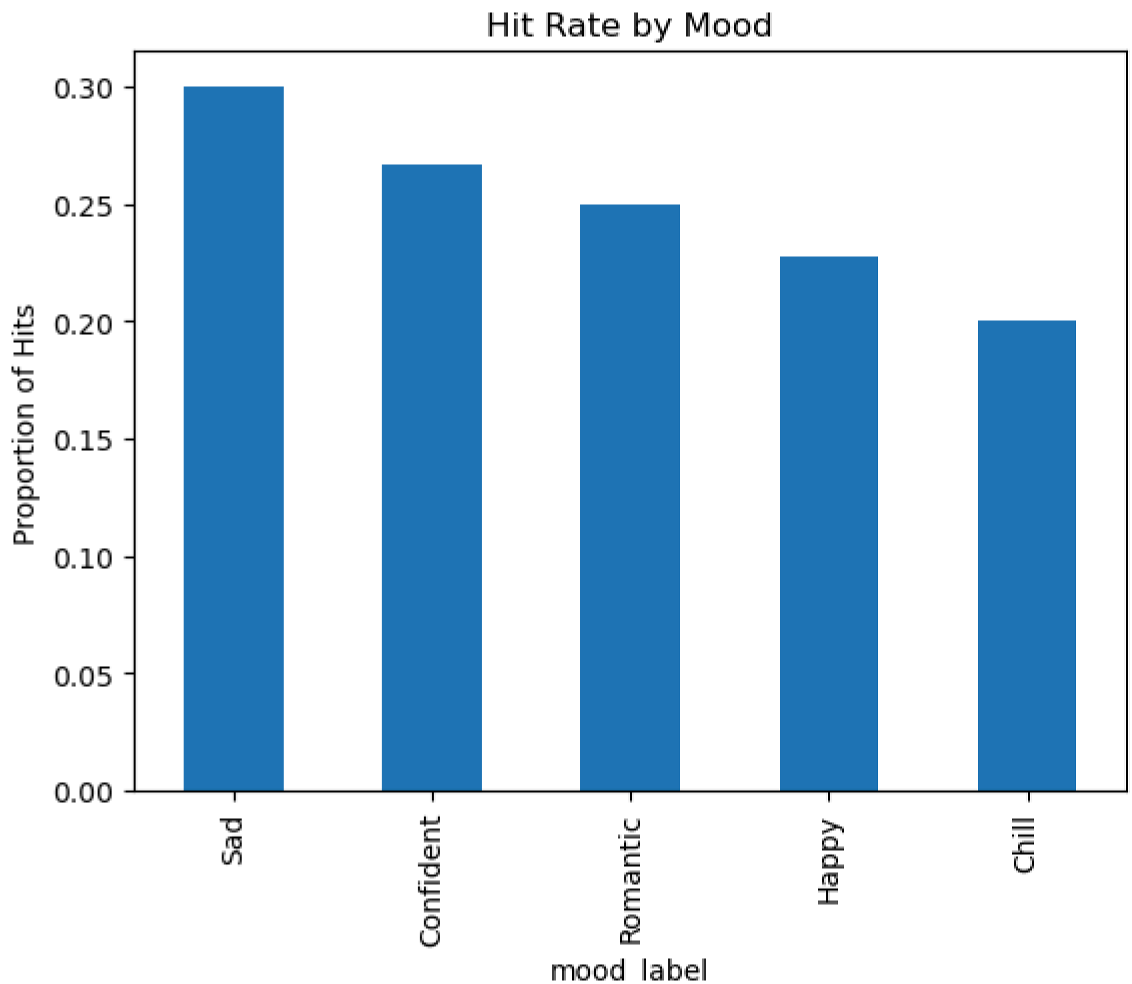
In [44]:
```
sns.countplot(data=df, x='genre', hue='mood_label')
plt.title("Mood Distribution by Genre")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

#Afrobeats aristists makes more confident songs generally, while amapiano artists makes more happy music.
```



In [45]:
```
import plotly.express as px

fig = px.scatter(
    df,
    x='streams_per_day',
    y='popularity',
    color='genre',
    hover_data=['track_name', 'mood'],
    title="🔥 Hit Predictor Results"
)
fig.show()
```

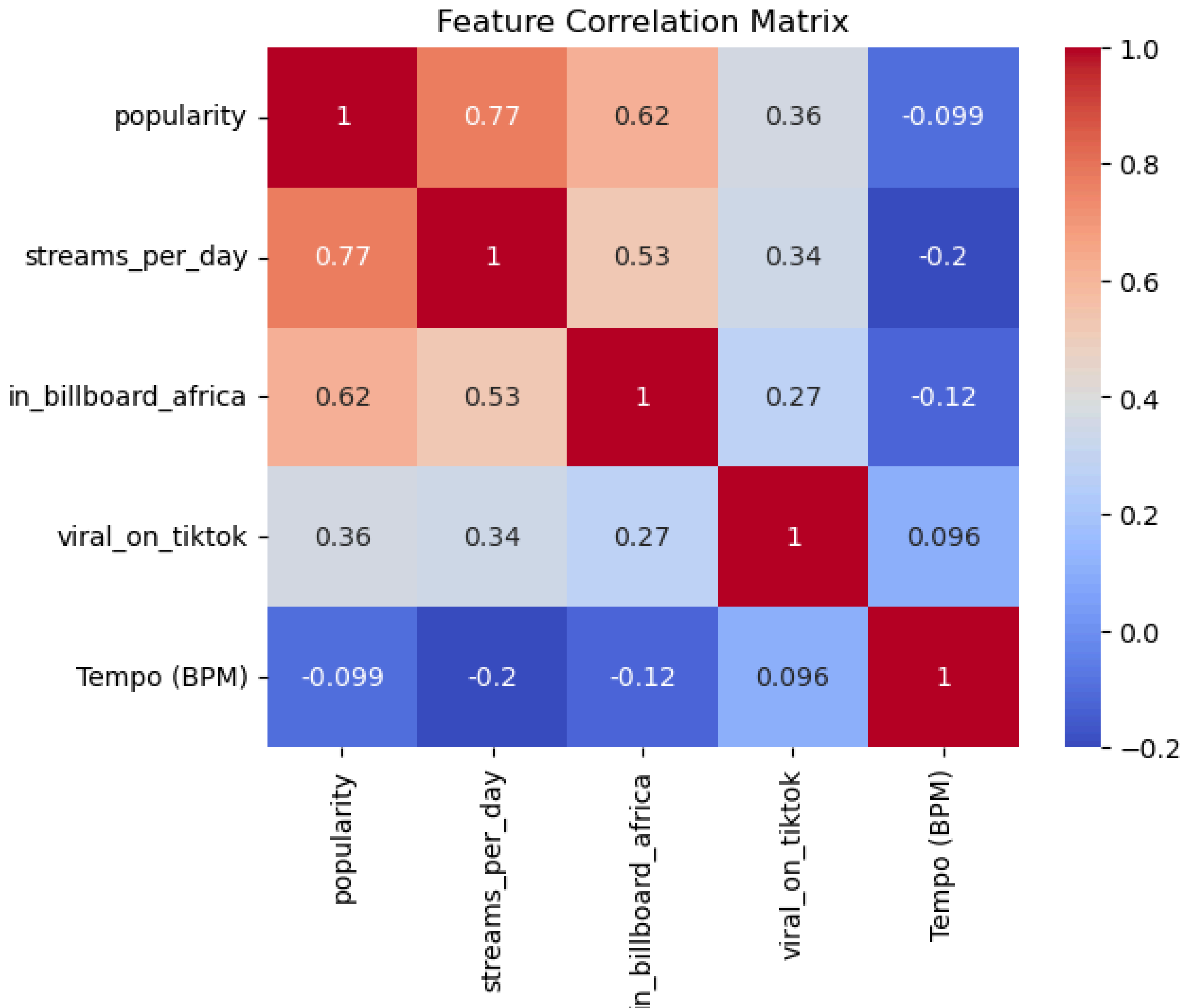


In [46]:
```
hit_rate_by_mood = df.groupby('mood_label')['is_hit'].mean().sort_values(ascending=False)
hit_rate_by_mood.plot(kind='bar', title="Hit Rate by Mood")
plt.ylabel("Proportion of Hits")
plt.show()

#Sad and confident songs are more likely to be hits (without seperating genres)
```



In [47]:

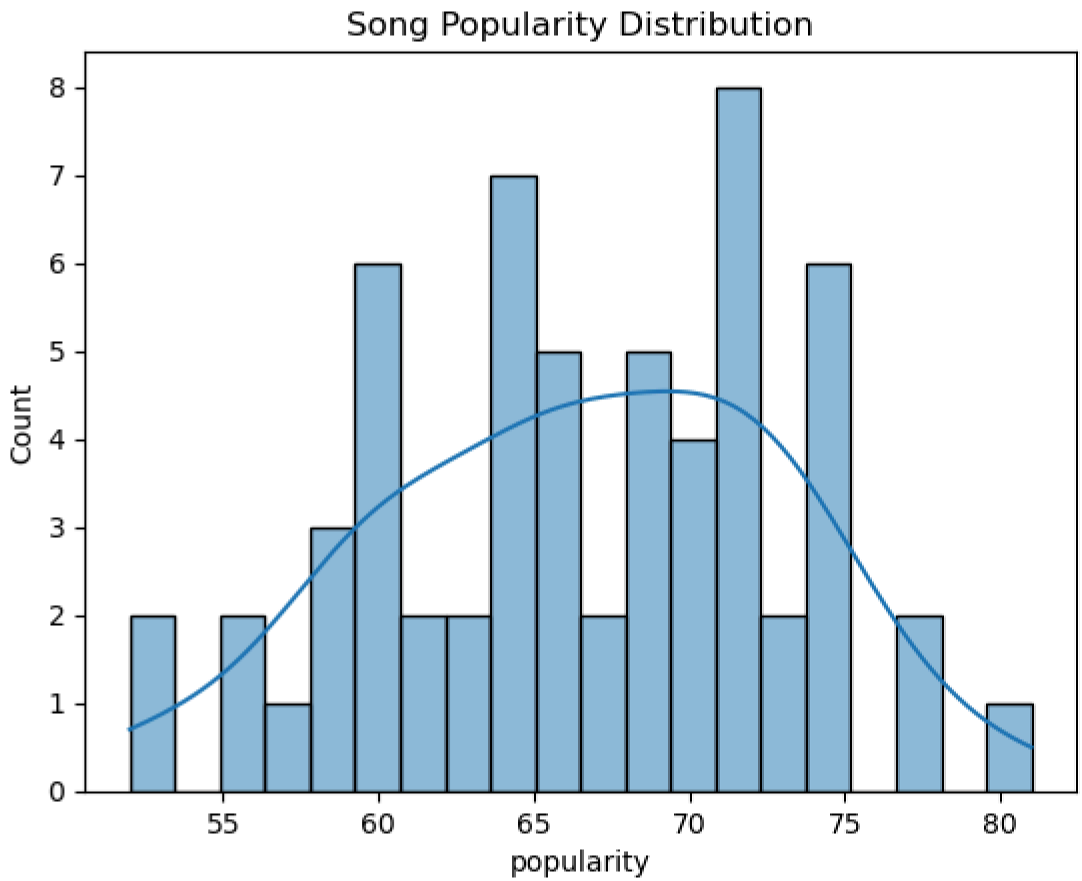
```
corr = df[['popularity', 'streams_per_day', 'in_billboard_africa', 'viral_on_tiktok', 'Tempo (BPM)']].corr()
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Feature Correlation Matrix")
plt.show()

#The correlation matrix shows strong alignment between popularity and streams per day,
#while TikTok virality and tempo remain largely independent.
#This suggests minimal redundancy and supports keeping all features in the model.
```



In [48]:

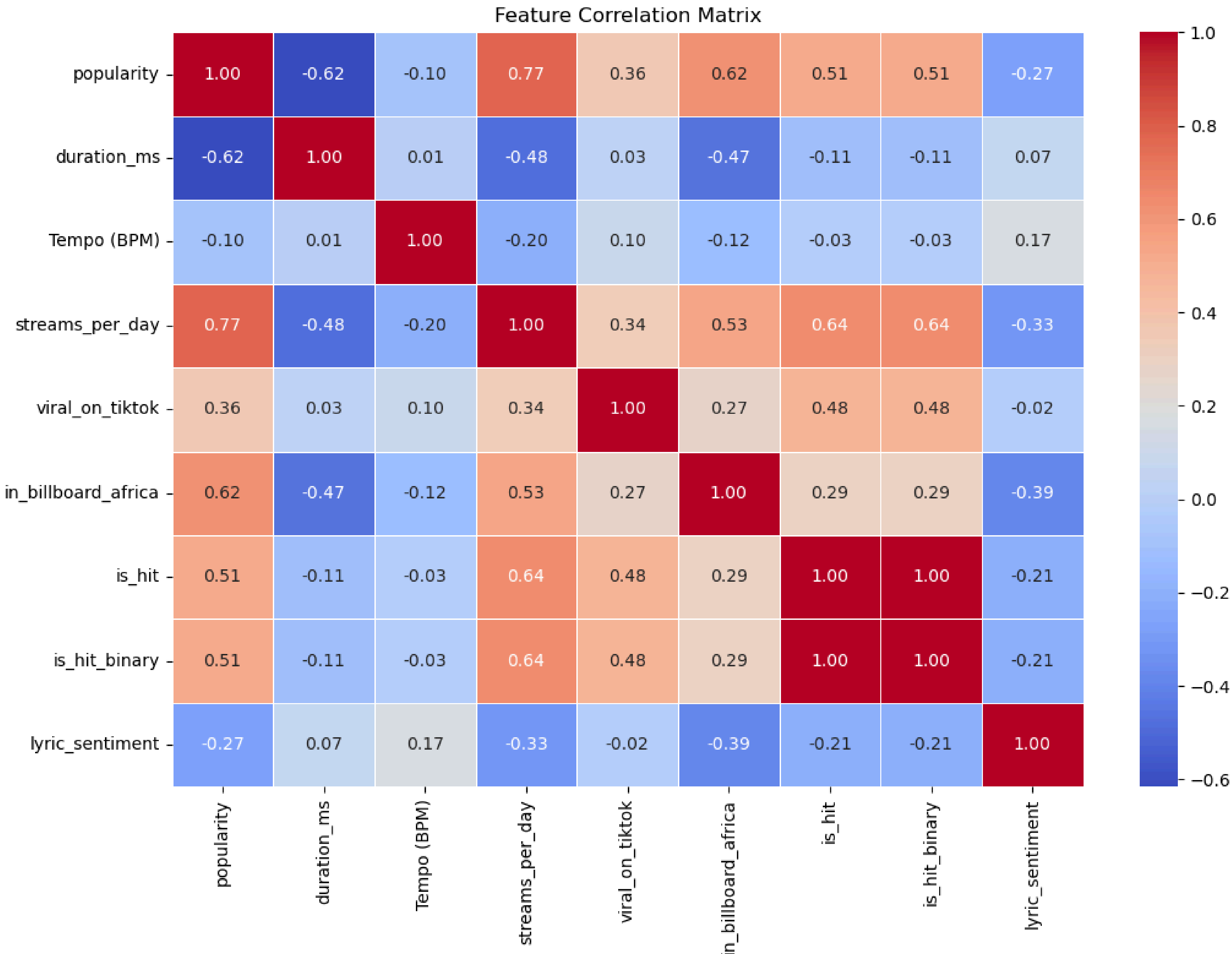
```
import seaborn as sns
import matplotlib.pyplot as plt

#Popularity distribution
sns.histplot(df['popularity'], bins=20, kde=True)
plt.title('Song Popularity Distribution')
plt.show()

#Visualizes the distribution of song popularity scores in the dataset.
```

```
#The distribution appears slightly right-skewed, with most songs clustering between 60 and 75.
#This helps inform threshold decisions when defining what constitutes a "popular" or "hit" song.
```

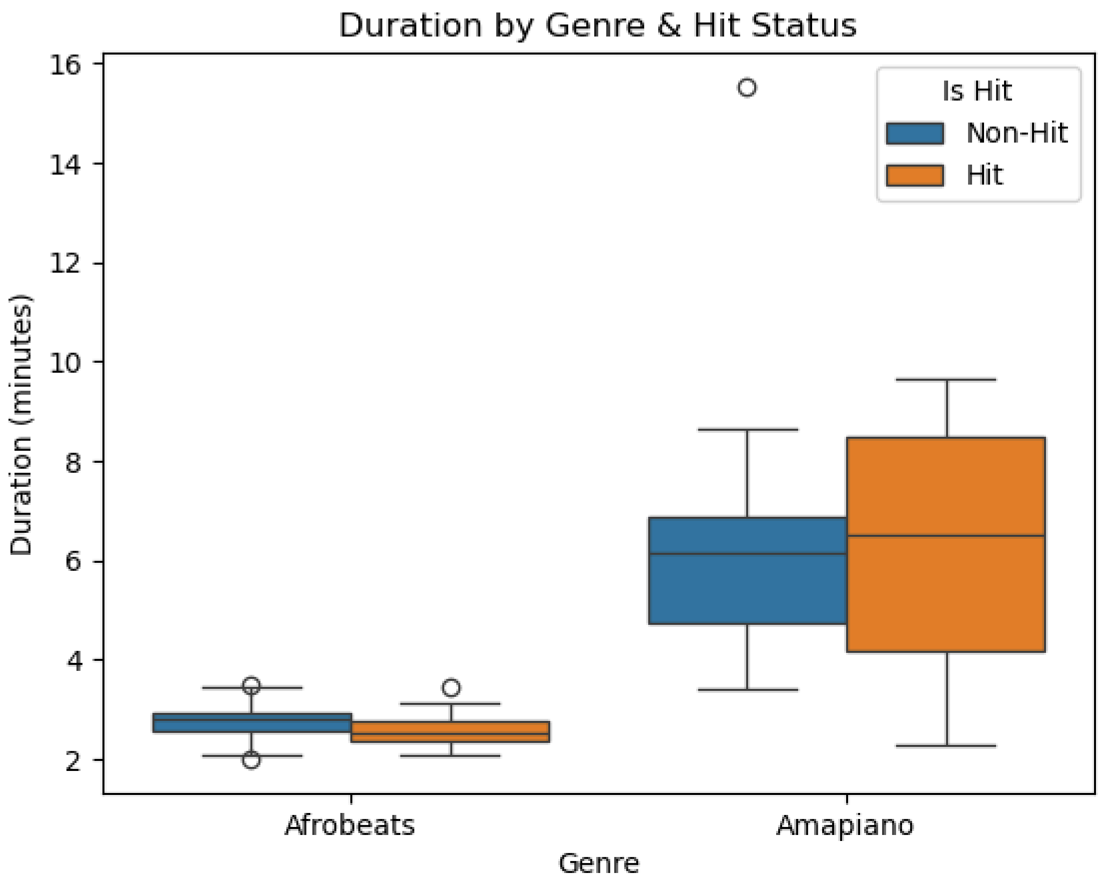


In [49]:
```
import seaborn as sns
import matplotlib.pyplot as plt

#Compute correlation matrix
correlation_matrix = df.corr(numeric_only=True)

#Plot heatmap
plt.figure(figsize=(12, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)
plt.title('Feature Correlation Matrix')
plt.show()

#Comprehensive correlation matrix for all numeric features.
```

```
#Popularity, streams_per_day, and in_billboard_africa show strong positive correlations with is_hit.
#Viral_on_tiktok is moderately correlated with hit status, while tempo and duration show weak or no correlation.
#Supports feature inclusion and confirms no severe multicollinearity.
```

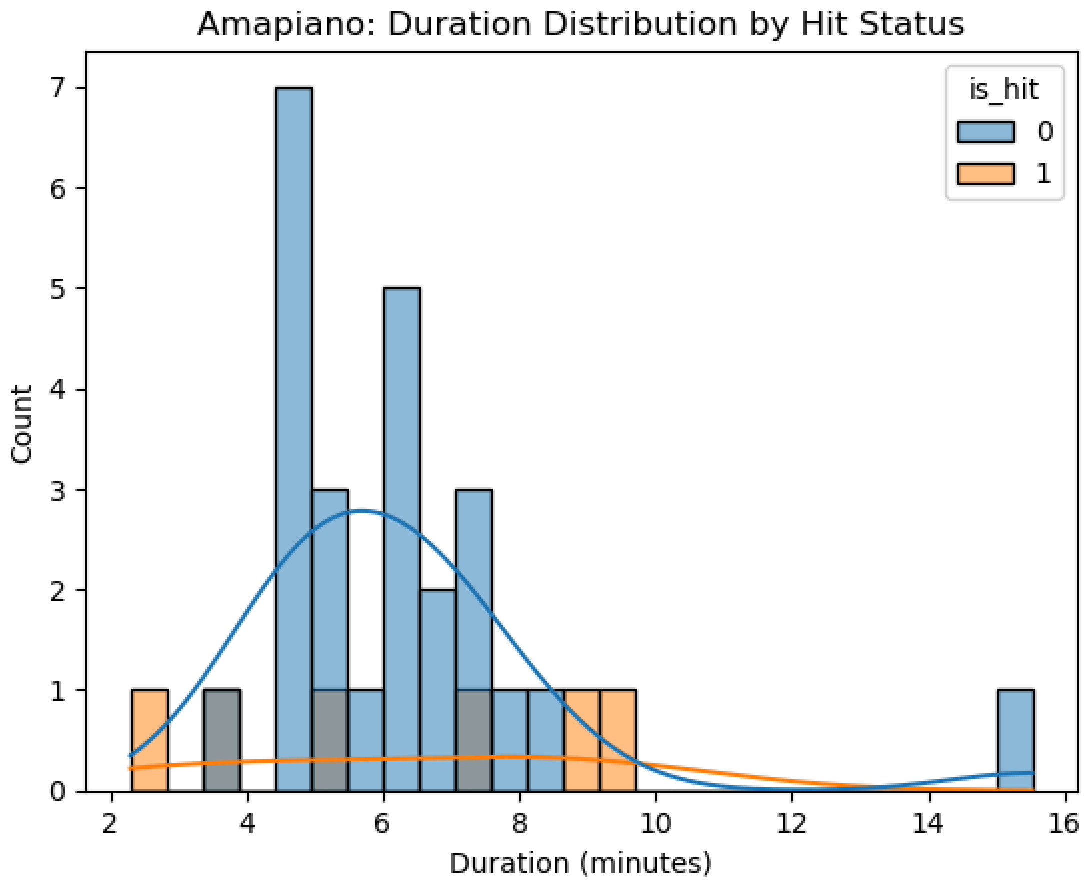

Feature Correlation Matrix
popularity
duration_ms
Tempo (BPM)
streams_per_day
viral_on_tiktok
in_billboard_africa
is_hit
is_hit_binary
lyric_sentiment
1.00 -0.62 -0.10 0.77 0.36 0.62 0.51 0.51 -0.27
-0.62 1.00 0.01 -0.48 0.03 -0.47 -0.11 -0.11 0.07
-0.10 0.01 1.00 -0.20 0.10 -0.12 -0.03 -0.03 0.17
0.77 -0.48 -0.20 1.00 0.34 0.53 0.64 0.64 -0.33
0.36 0.03 0.10 0.34 1.00 0.27 0.48 0.48 -0.02
0.62 -0.47 -0.12 0.53 0.27 1.00 0.29 0.29 -0.39
0.51 -0.11 -0.03 0.64 0.48 0.29 1.00 1.00 -0.21
0.51 -0.11 -0.03 0.64 0.48 0.29 1.00 1.00 -0.21
-0.27 0.07 0.17 -0.33 -0.02 -0.39 -0.21 -0.21 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

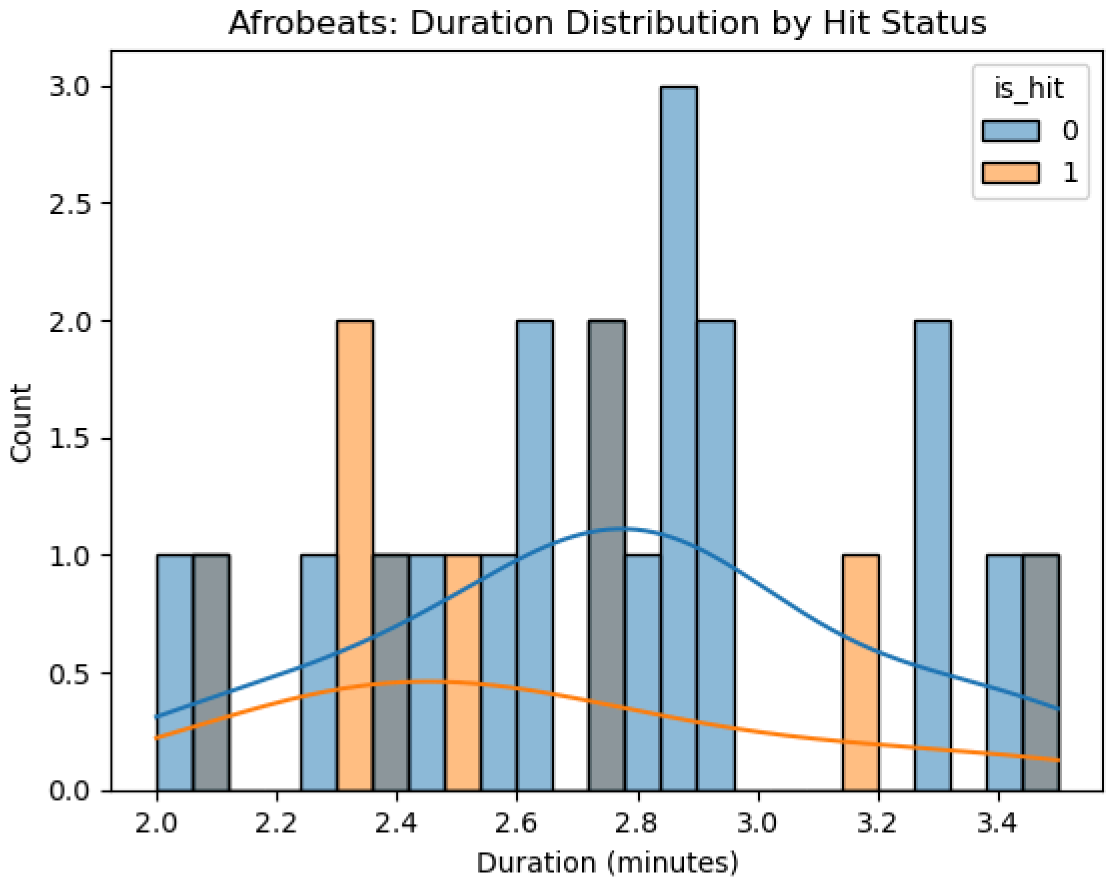
In [50]:
```
#Convert duration from milliseconds to minutes
df['duration_min'] = df['duration_ms'] / 60000

#Map 0/1 to readable labels
df['hit_label'] = df['is_hit'].map({0: 'Non-Hit', 1: 'Hit'})

#Boxplot grouped by genre and hit status
sns.boxplot(x='genre', y='duration_min', hue='hit_label', data=df)
plt.title('Duration by Genre & Hit Status')
plt.xlabel('Genre')
plt.ylabel('Duration (minutes)')
plt.legend(title='Is Hit')
plt.show()


#Amapiano songs tend to be longer overall, with hit songs slightly longer on average than non-hits.
#Afrobeats songs show less variation in duration, and hits appear slightly shorter.
```



In [51]:
```
#Histogram for Each Genre Separately for duration_ms
#Amapiano duration distribution
sns.histplot(data=df[df['genre'] == 'Amapiano'], x='duration_min', hue='is_hit', bins=25, kde=True)
plt.title('Amapiano: Duration Distribution by Hit Status')
plt.xlabel('Duration (minutes)')
plt.show()

#Afrobeats duration distribution
sns.histplot(data=df[df['genre'] == 'Afrobeats'], x='duration_min', hue='is_hit', bins=25, kde=True)
plt.title('Afrobeats: Duration Distribution by Hit Status')
plt.xlabel('Duration (minutes)')
plt.show()

#Histograms of song duration by hit status, split by genre.
#Amapiano hits span a wider range and tend to be longer than non-hits.
```

```
#Afrobeats durations are more tightly clustered, with hits appearing slightly shorter
#These plots help visualize how duration influences hit potential differently across genres.
```





10. Conclusion & Future Work

This project applied machine learning to analyze and predict hit songs across Afrobeats and Amapiano genres, combining streaming data, audio features, chart performance, and virality metrics. High **Spotify popularity** and **streaming velocity** were strongly associated with hits, but the analysis suggests these metrics behave more as **symptoms** of success than as causes—they describe a song that has *already* broken out rather than explaining why it did.

In contrast, commonly assumed musical drivers of success—**tempo**, **duration**, and **beat strength**—showed minimal predictive power (near-zero correlation with hit status in the exploratory analysis). This supports the idea that, in today's digital music ecosystem, a song's **shareability and visibility** often matter more than its audio structure. Audio features were still useful *descriptively*: Afrobeats hits leaned toward **lower tempos**, while Amapiano hits clustered around **consistent mid-tempo** ranges—patterns that diverge from typical U.S. pop hit profiles.

When **lyric sentiment** (Genius + VADER) was added as a feature, it ranked among the top predictors for Afrobeats—above TikTok virality and Billboard presence—but it **did not improve hit detection**, and slightly lowered the Afrobeats hit-class F1 (with no change for Amapiano). On a dataset this small (~30 songs per genre, few labelled hits), a one-song shift moves F1 substantially, so this is best read as noise rather than evidence the feature is harmful. The likeliest cause is **VADER's English-only training**: it cannot meaningfully score Amapiano's Zulu/Xhosa lyrics, and missing scores were imputed with the genre median. The signal is promising for Afrobeats but unproven—more labelled songs and proper multilingual sentiment scoring are needed.

A key limitation is label leakage. Because `is_hit` is defined partly from `streams_per_day` and `popularity`, the high importance of those two features is somewhat circular—the model is partly predicting hits from the same signals used to define them. The more meaningful result is that **non-defining features** like TikTok virality and Billboard presence outranked musical attributes such as beat strength and tempo, suggesting that success is driven more by distribution and reach than by a song's sonic characteristics. A stronger test of true predictability would forecast hits using only song features and early virality signals, excluding the streaming metrics that overlap with the label.

The best-performing model was a **Random Forest classifier with class weighting**, evaluated through 5-fold stratified cross-validation. Reporting the **hit (minority) class**, it achieved:

- **Afrobeats: F1 0.94** (precision 1.00, recall 0.89)
- **Amapiano: F1 0.91** (precision 1.00, recall 0.83)

These reflect perfect precision and strong recall on the hit class, and they outperform genre-specific logistic regression (hit-class F1 ≈ 0.78 for Afrobeats and ≈ 0.83 for Amapiano)—confirming that the Random Forest is markedly better at detecting the minority hit class. (This is the model *without* `lyric_sentiment`, which gave the strongest hit-class F1.)

Future Work

To further improve performance and expand applicability:

- **Improve lyric sentiment for non-English lyrics**, since many Amapiano tracks are in Zulu/Xhosa that VADER scores unreliably.
- **Test true predictive power without leakage** by forecasting hits from audio features and early virality only, excluding the streaming metrics used to define the label.
- **Analyze playlist placement, release timing, and artist reputation** as upstream exposure signals.
- **Extend the cross-genre modeling** to additional African subgenres for broader generalization.
- Explore **model stacking or gradient boosting** to capture nonlinear feature interactions.
- **Grow the labelled dataset**, since ~30 songs per genre limits how far any of these conclusions can be pushed.

These findings offer practical insights for artists, producers, and digital marketers aiming to understand or influence the trajectory of songs in the era of algorithmic culture.